

MATSE: Operations Research

Introduction

Lecturer: Barbara Franci

`b.franci@maastrichtuniversity.nl`

Office: Paul-Henri Spaaklaan 1, C4.011

Maastricht, the Netherlands

Course Information

When:

- Every Wednesday
- Lectures+ Exercises (3/4 hours every week)
- Till December 4th + January 15th

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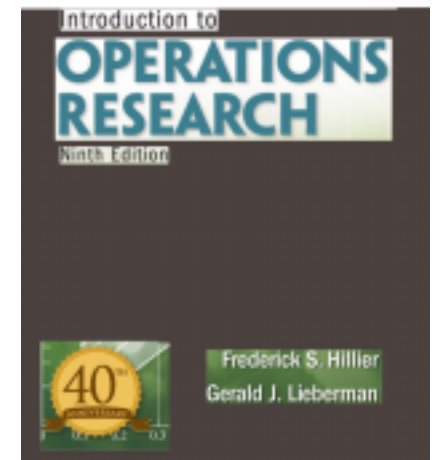
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Textbook

- Introduction to Operations Research by Hillier and Lieberman (9th ed)



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- You will have an impression of several facets of operations research
- Understanding the concepts is more important than memorization
- We all tend to forget the precise details, but my aim and hope is for you to learn the concepts that will allow you to relocate the details as and when you need them

Operations Research, a sample

The scope is much too broad for one course, but here we will sample some topics:

Deterministic Models:

Stochastic Models:

Game Theory

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- Linear Programming [HL: §3,4, and more!]
(Linear Algebra but looking for a *best* feasible solution)
- Transportation problem [HL: §8]

Stochastic Models:

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Deterministic Models:

- Linear Programming [HL: §3,4, and more!]
(Linear Algebra but looking for a *best* feasible solution)
- Transportation problem [HL: §8]

Stochastic Models:

- Coin flipping and more
- Markov Chains [HL: §16]
- Queuing Theory [HL: §17]

Game Theory

Introduce Yourself!

- Who
- Why
- Word

Introduce Yourself!

- Barbara Franci, Assistant Professor
- Degree in Math, Math is Fun!
- Ich spreche kein Deutsch



MATSE: Operations Research

Lecture 1. (linear) Deterministic Models

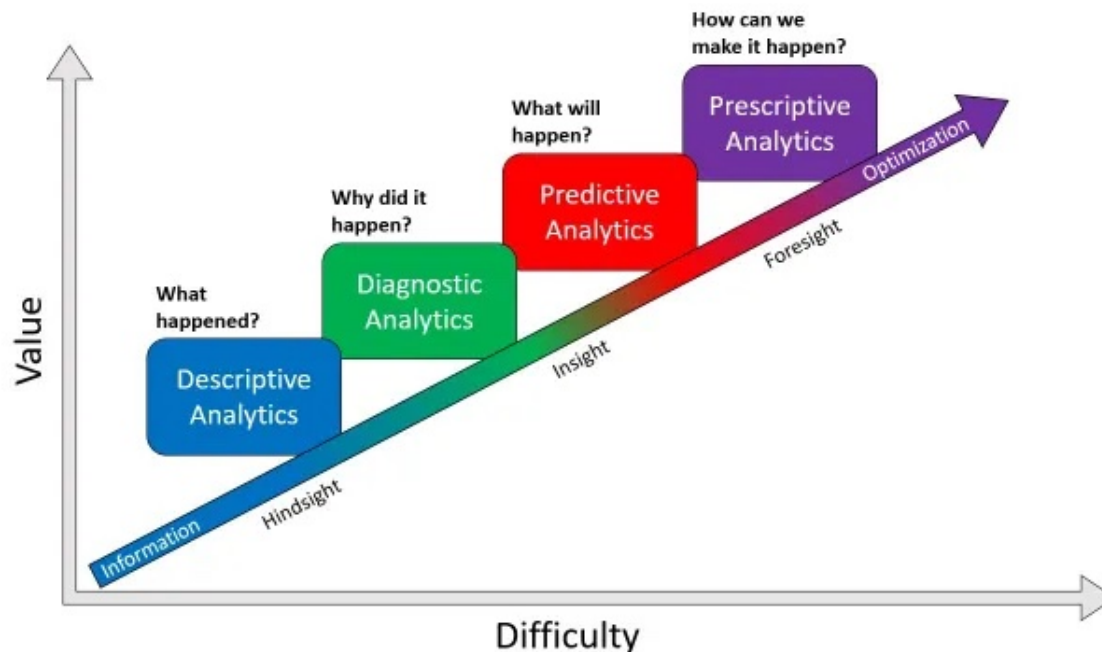
Why OR?

Operation Research is the discipline of applying advanced analytical methods to help **make better decisions**.

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Operation Research is the discipline of applying advanced analytical methods to help **make better decisions**.

Operation Research is **prescriptive**: what should we do in order to optimize performance subject to constraints?



<https://inthenameofai.com/data-analytics-not-data-analytics/>

Maximizing Gain

You are the boss of a small company, producing two products P_1 and P_2 . If you produce x_1 units of P_1 and x_2 units of P_2 , your gain is

$$G(x_1, x_2) = 30x_1 + 50x_2$$

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Your production runs on three machines M_A , M_B and M_C with the following capacities:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$

$$M_B: \quad x_1 + x_2 \leq 150$$

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Which choice of (x_1, x_2) maximizes your gain?

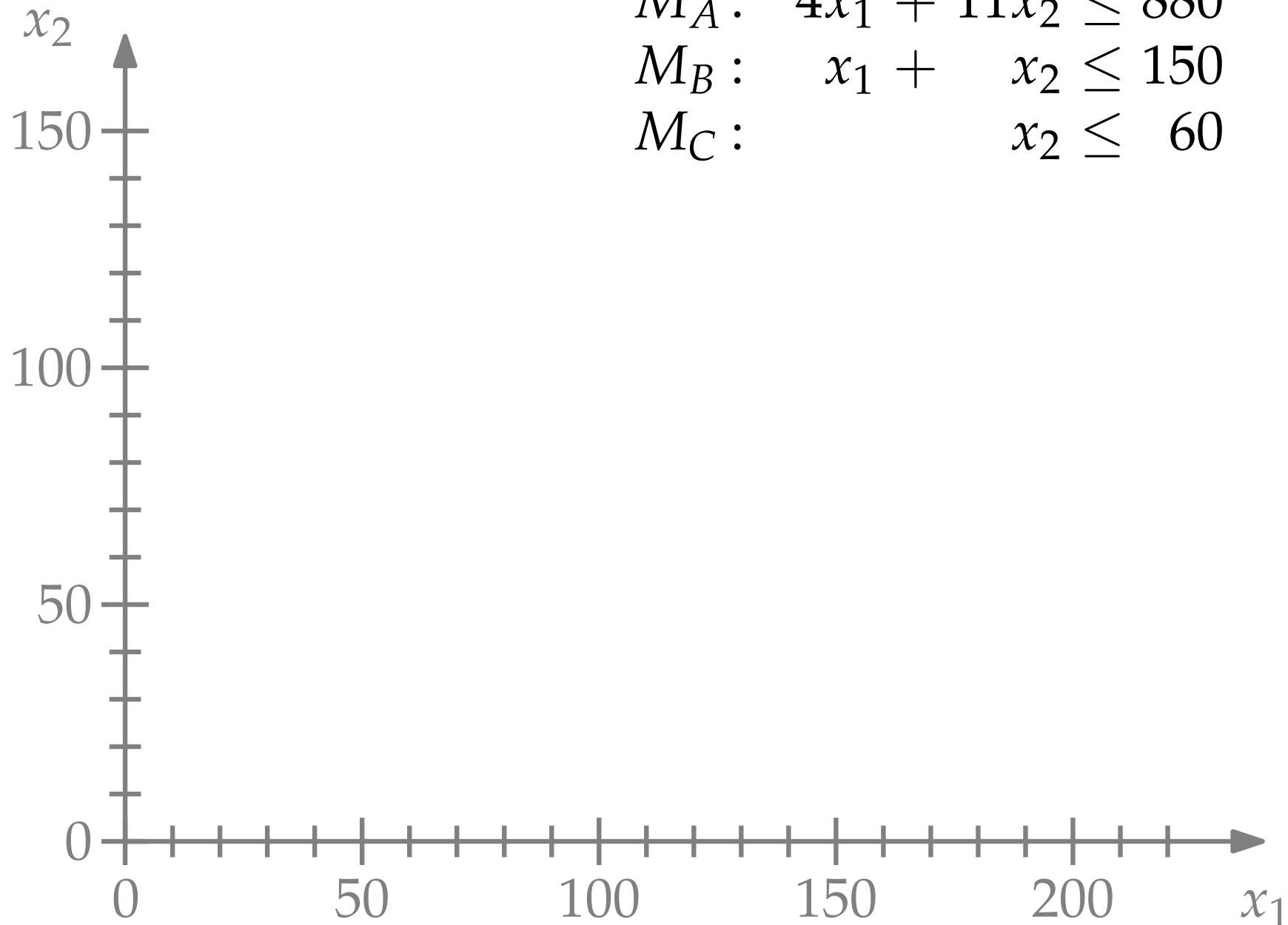
Solving by hand

Linear constraints:

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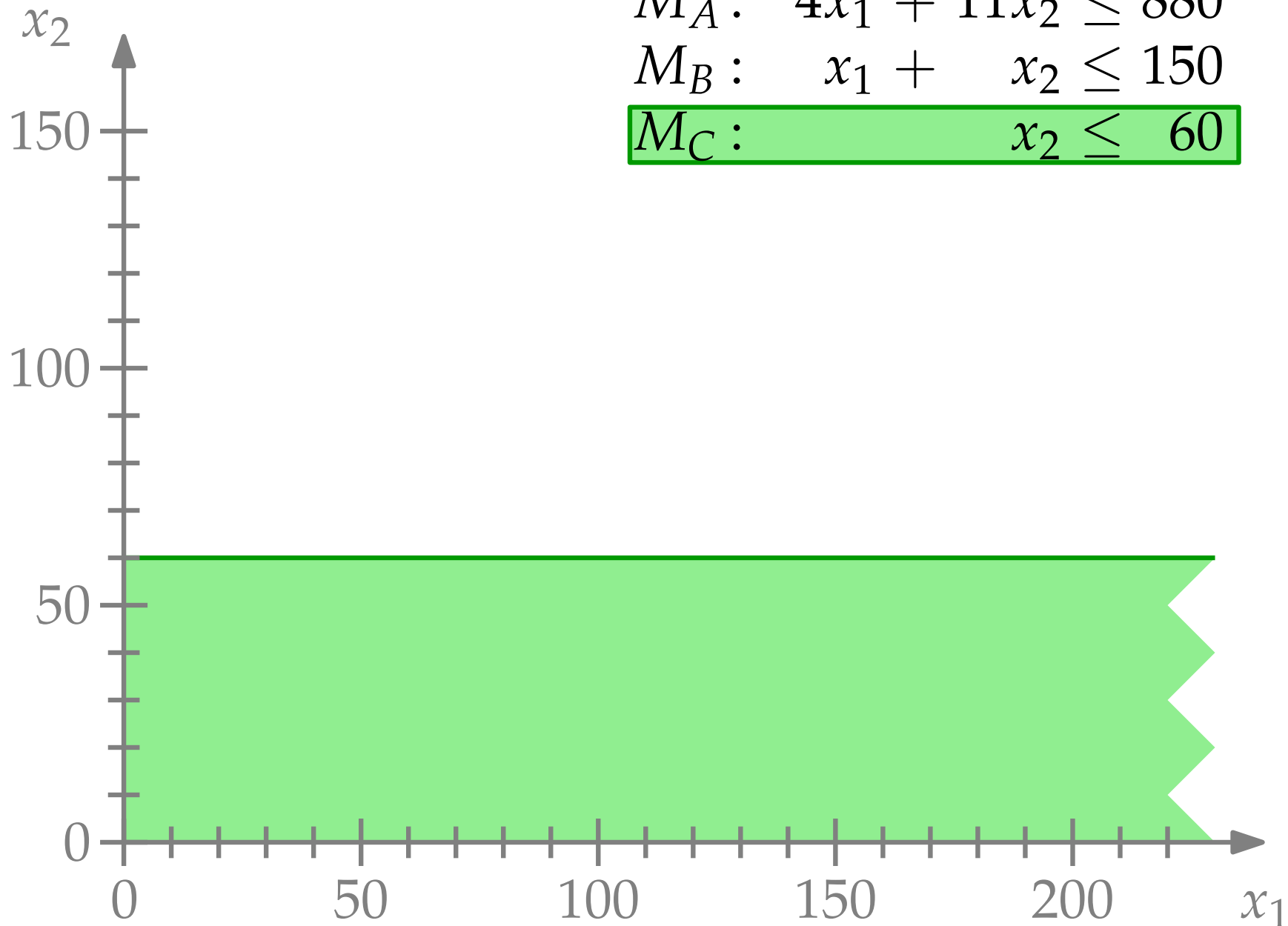
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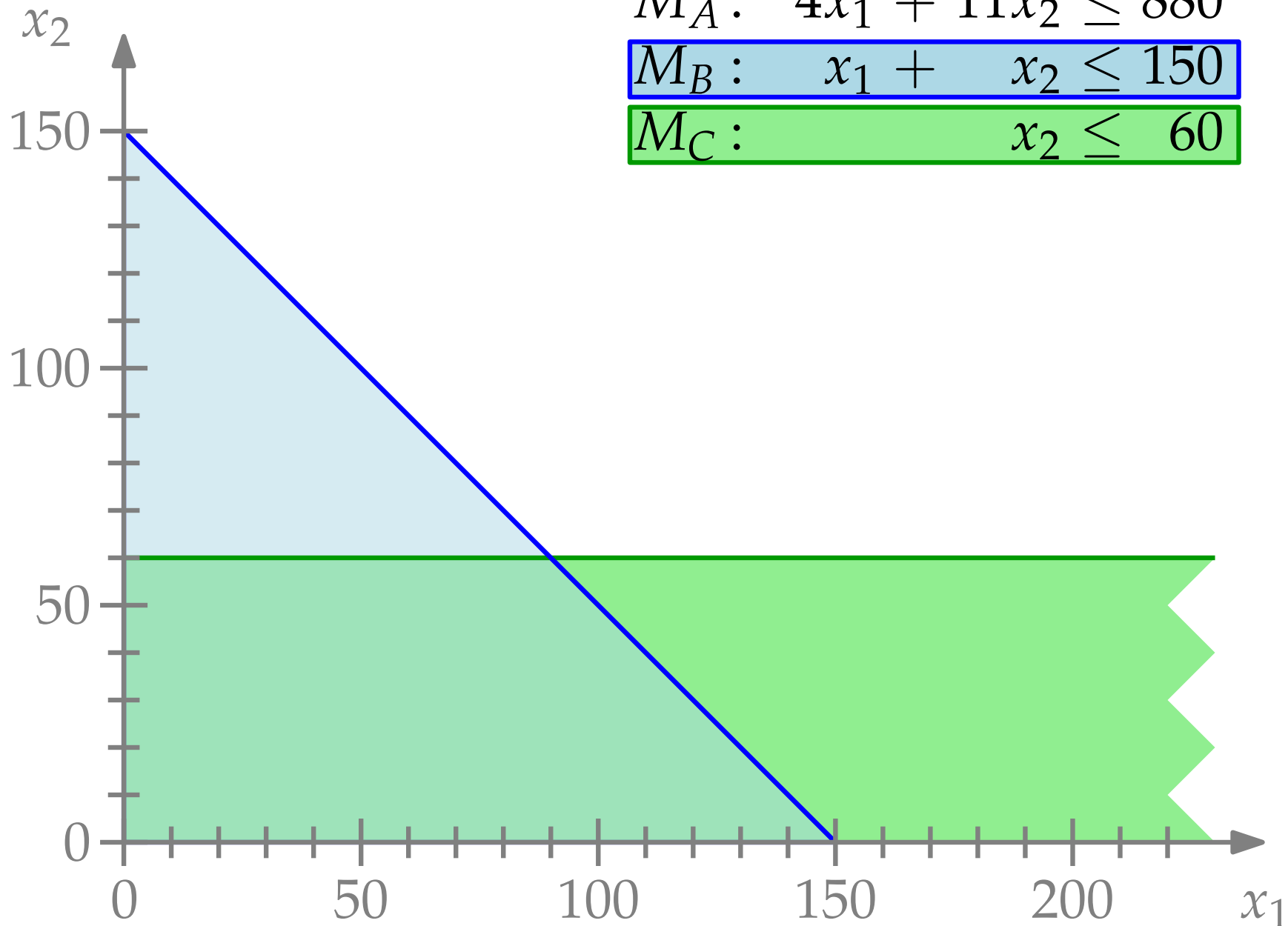
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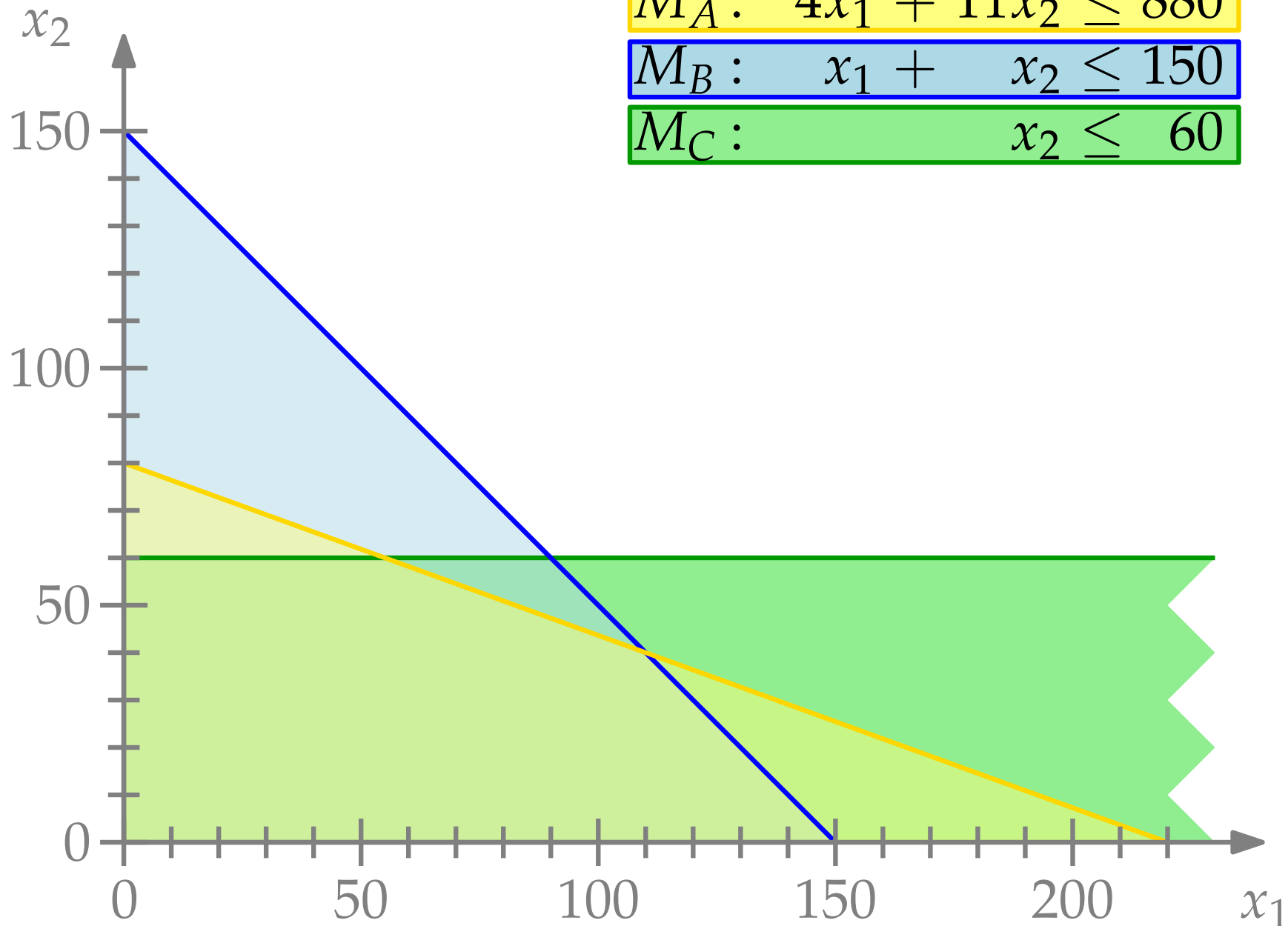
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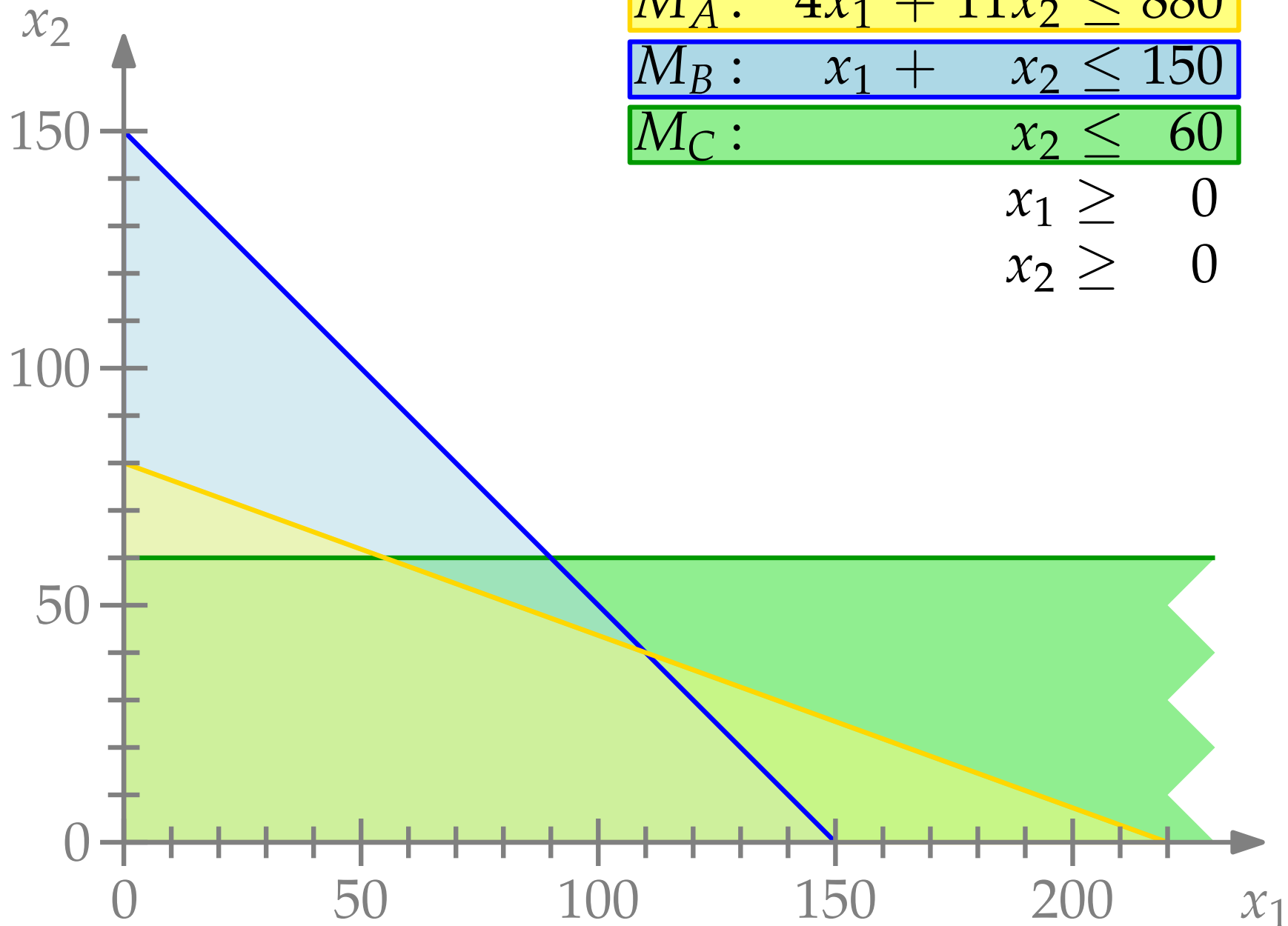
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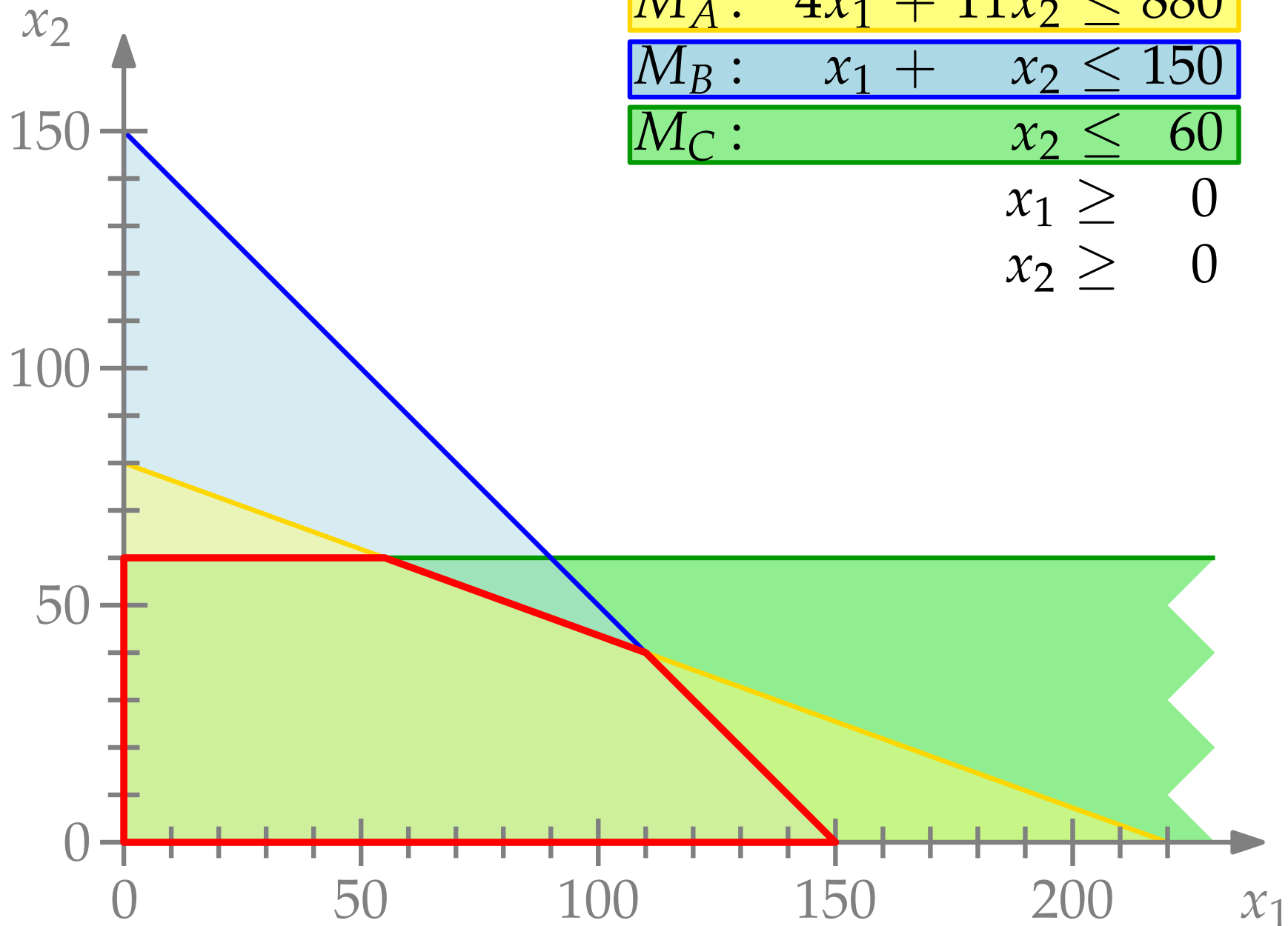
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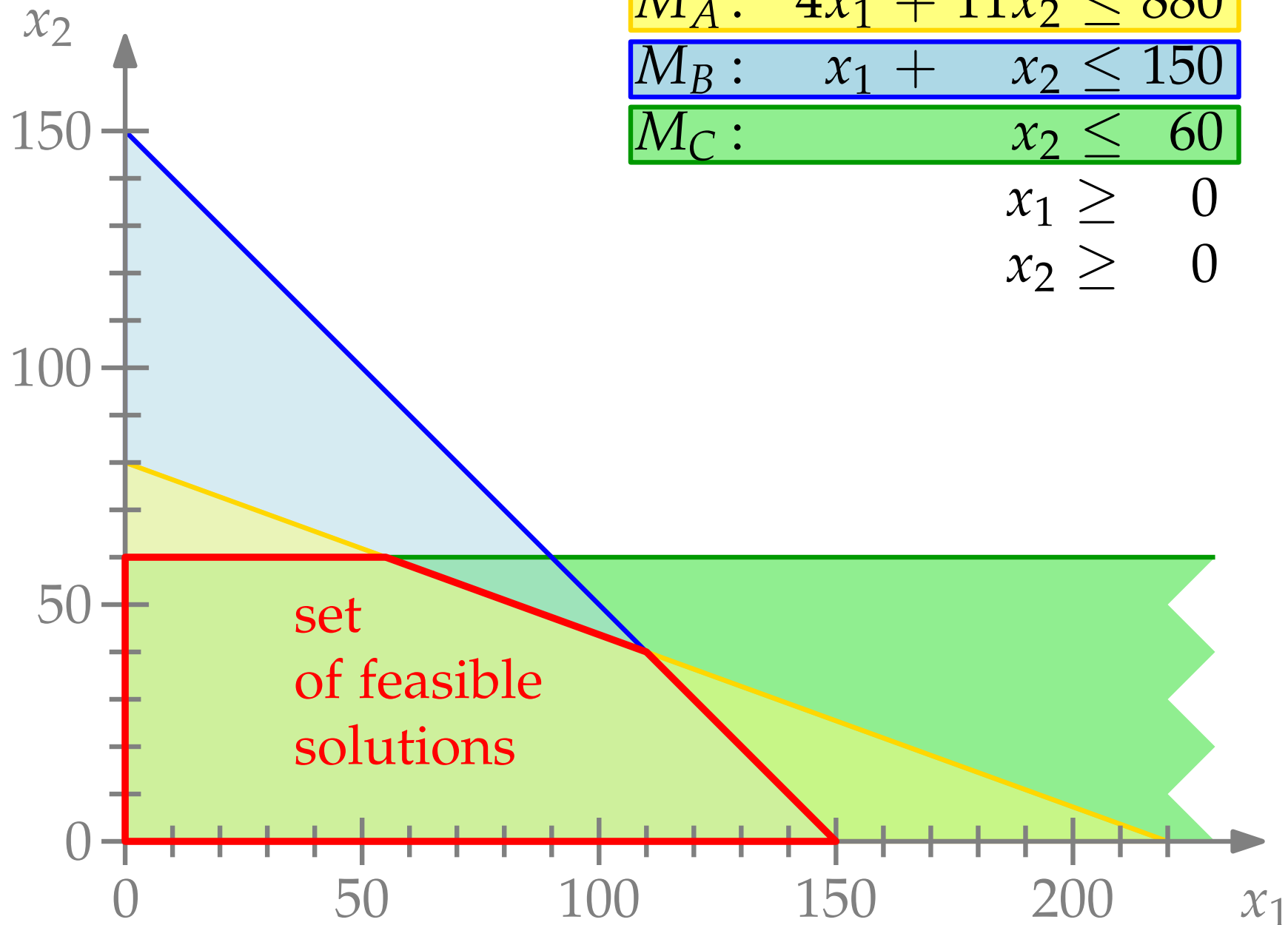
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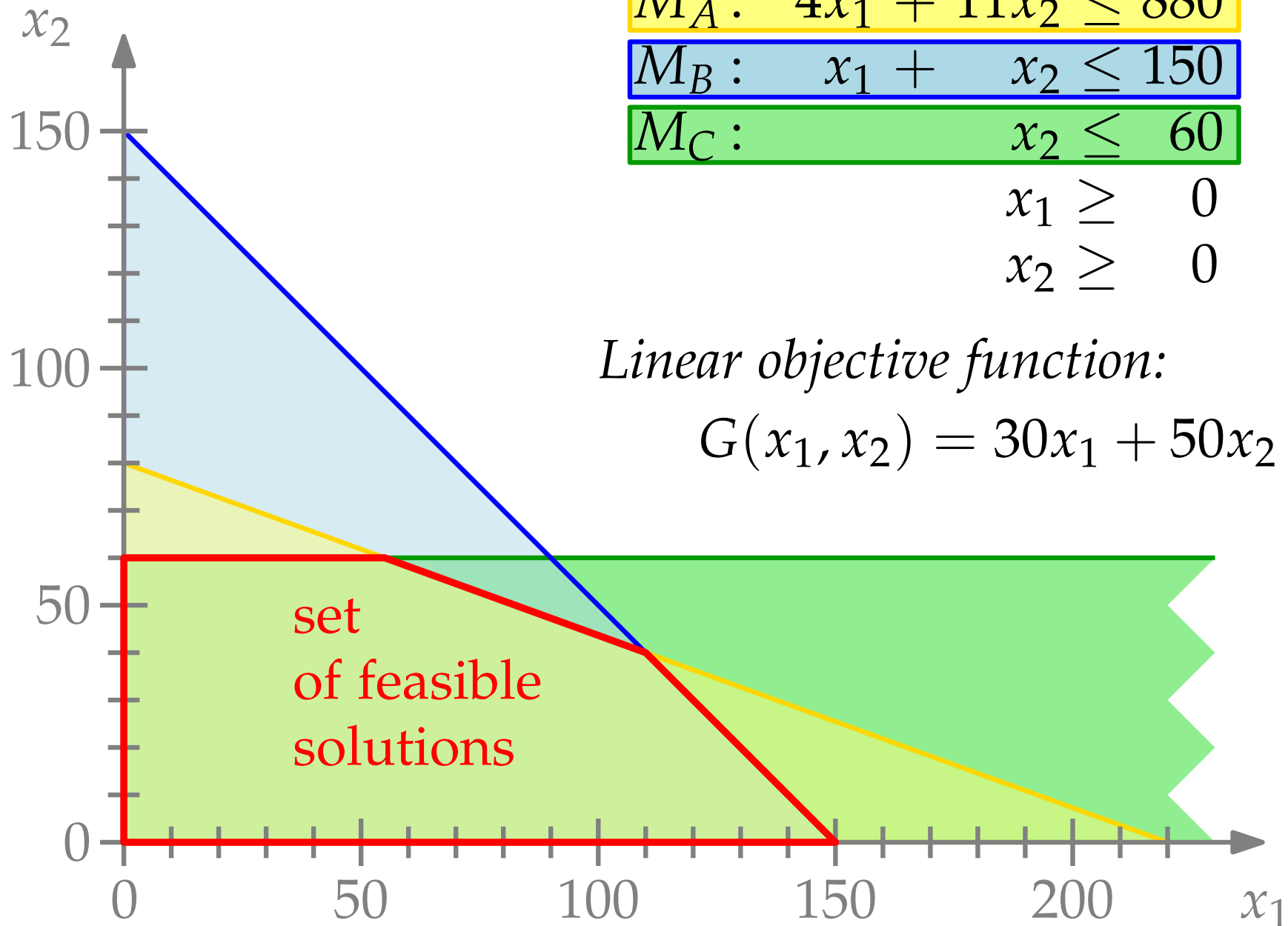
$$M_C: x_2 \leq 60$$

$$x_1 \geq 0$$

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Linear objective function:

$$G(x_1, x_2) = 30x_1 + 50x_2$$



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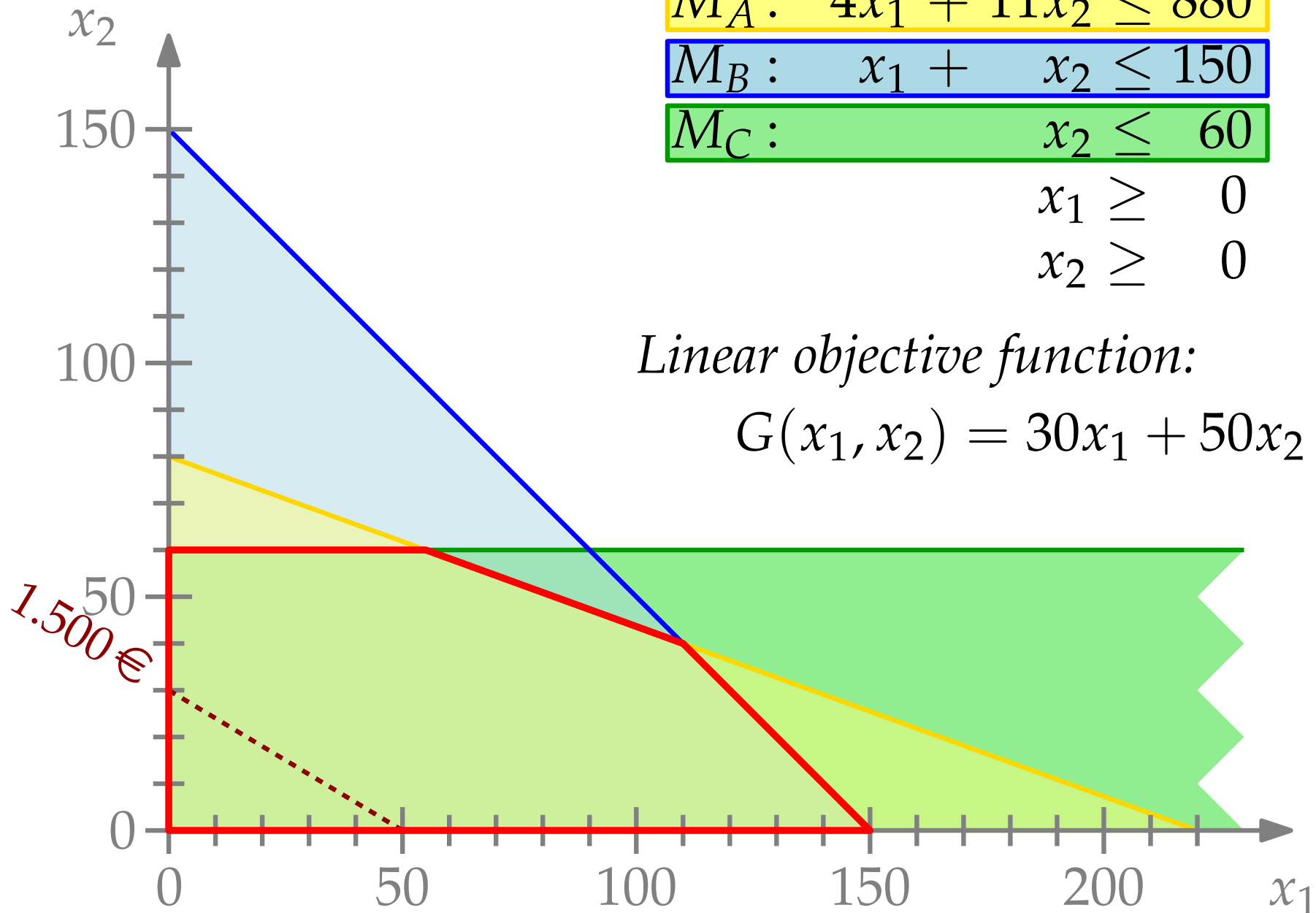
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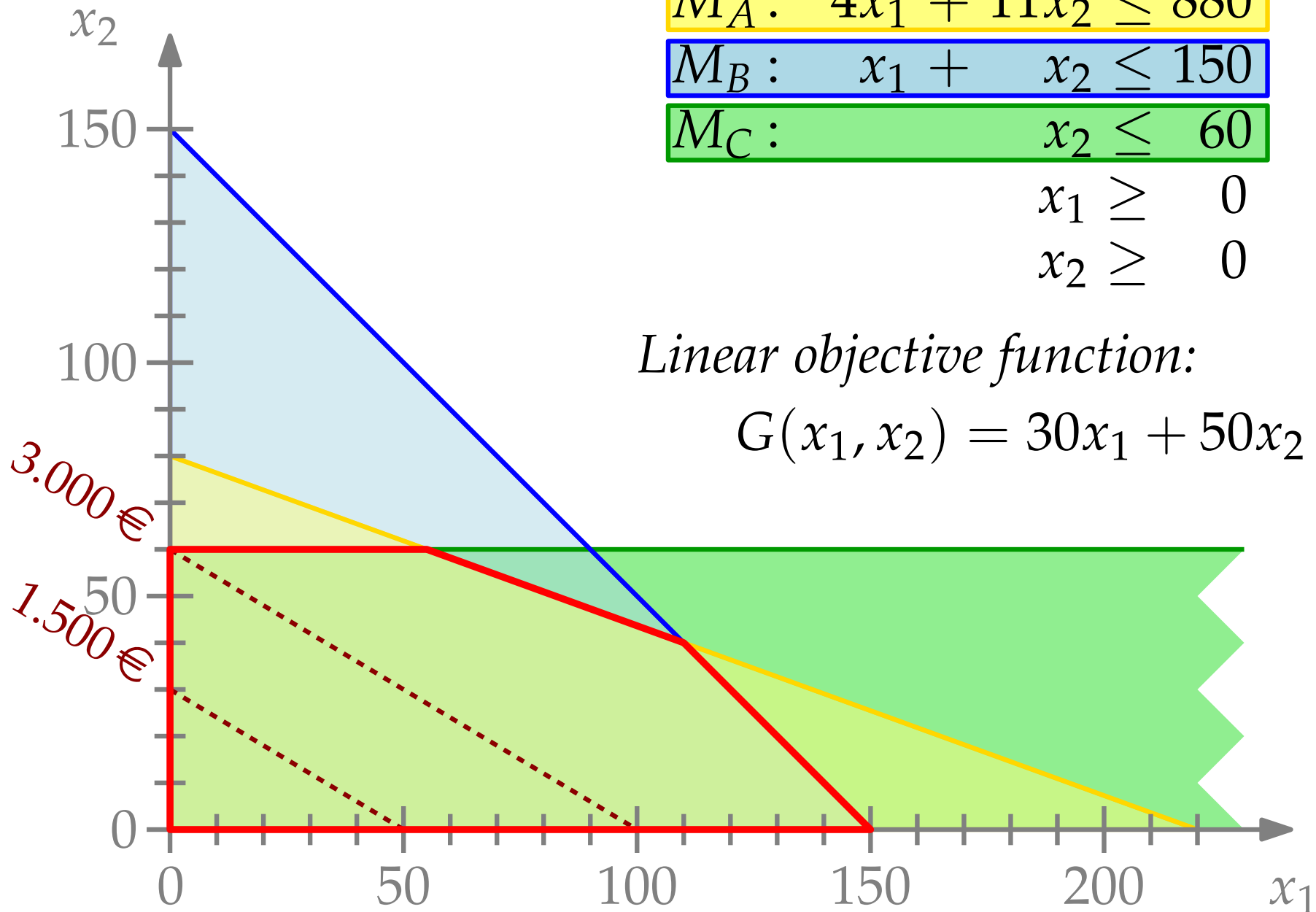
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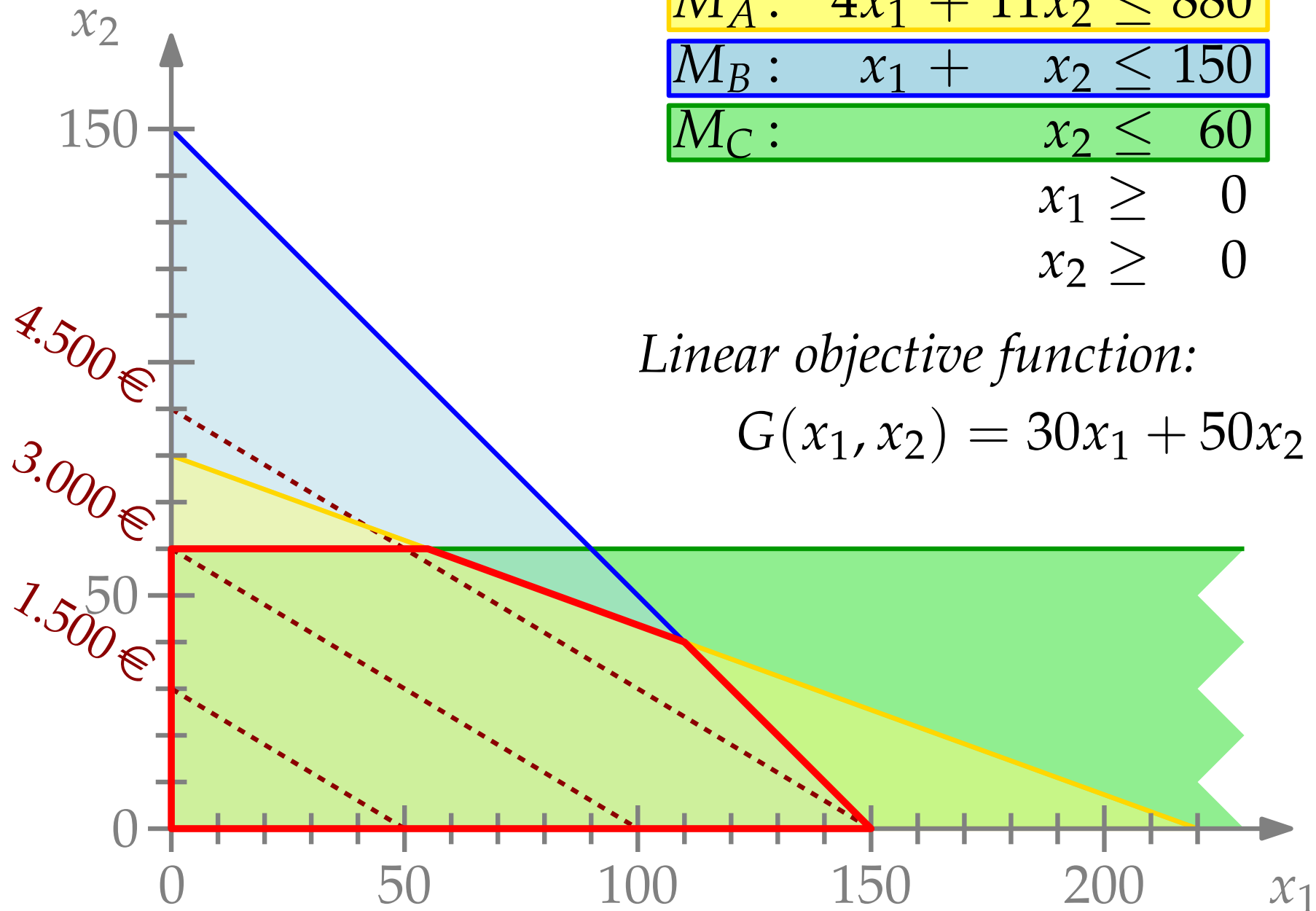
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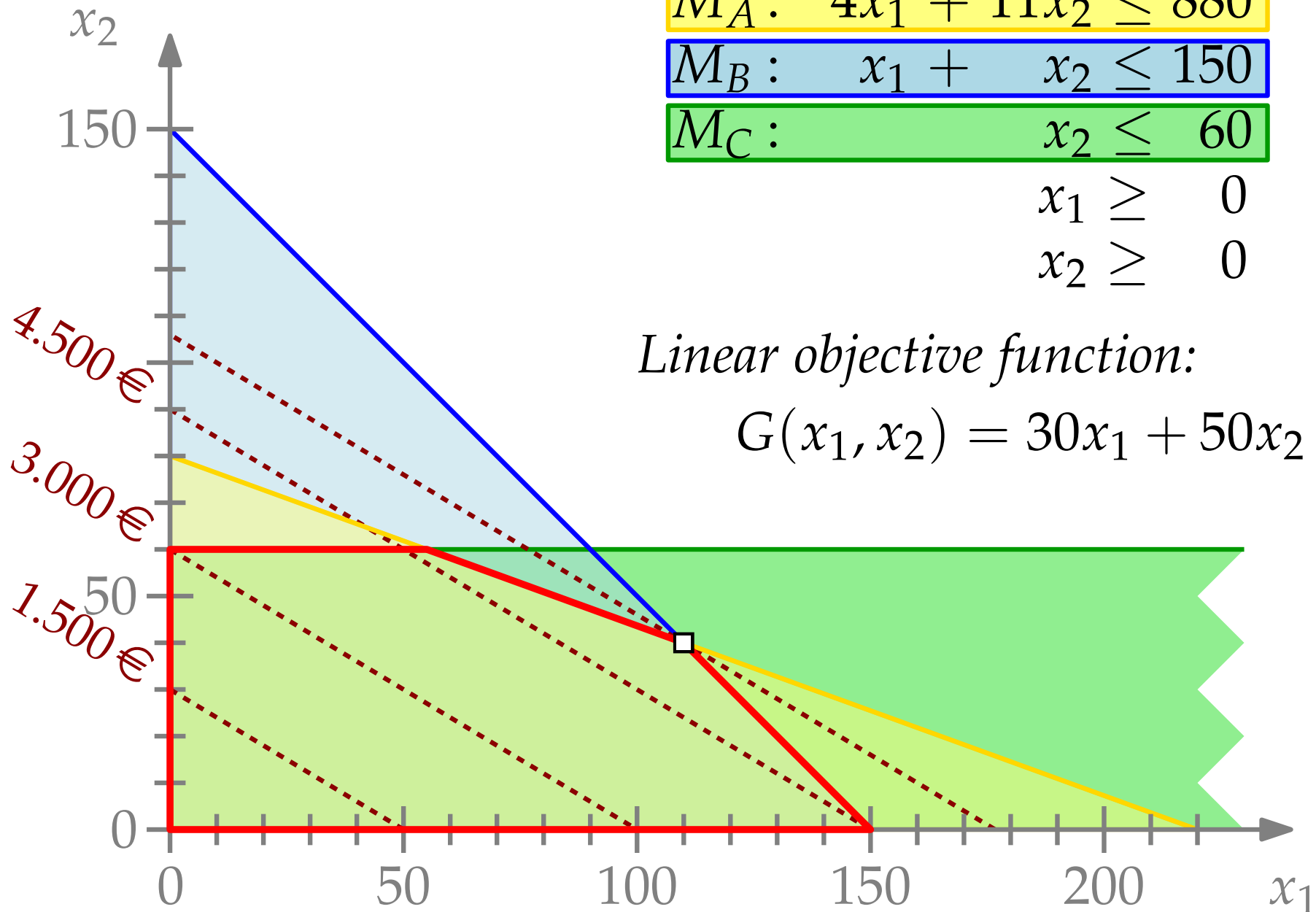
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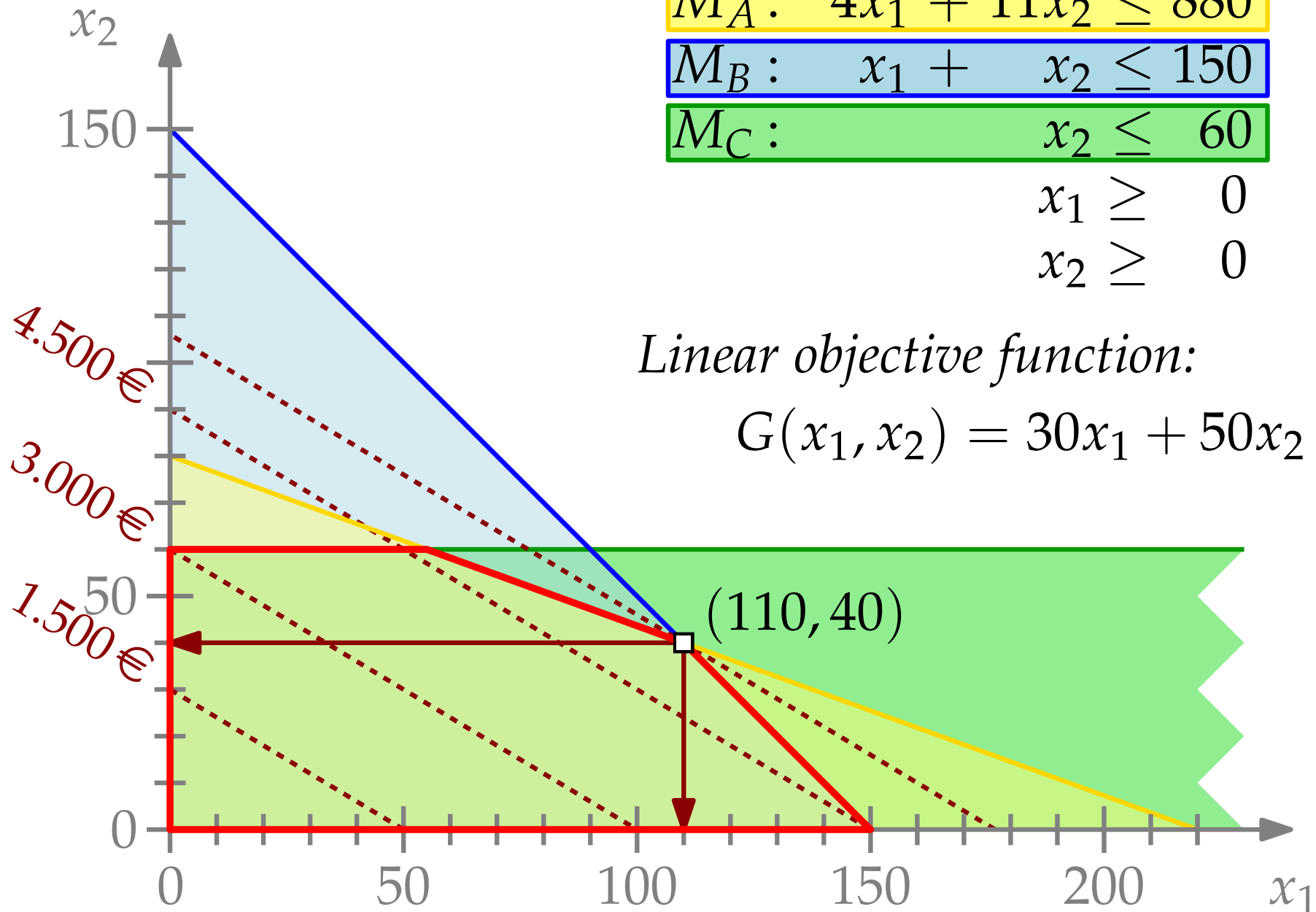
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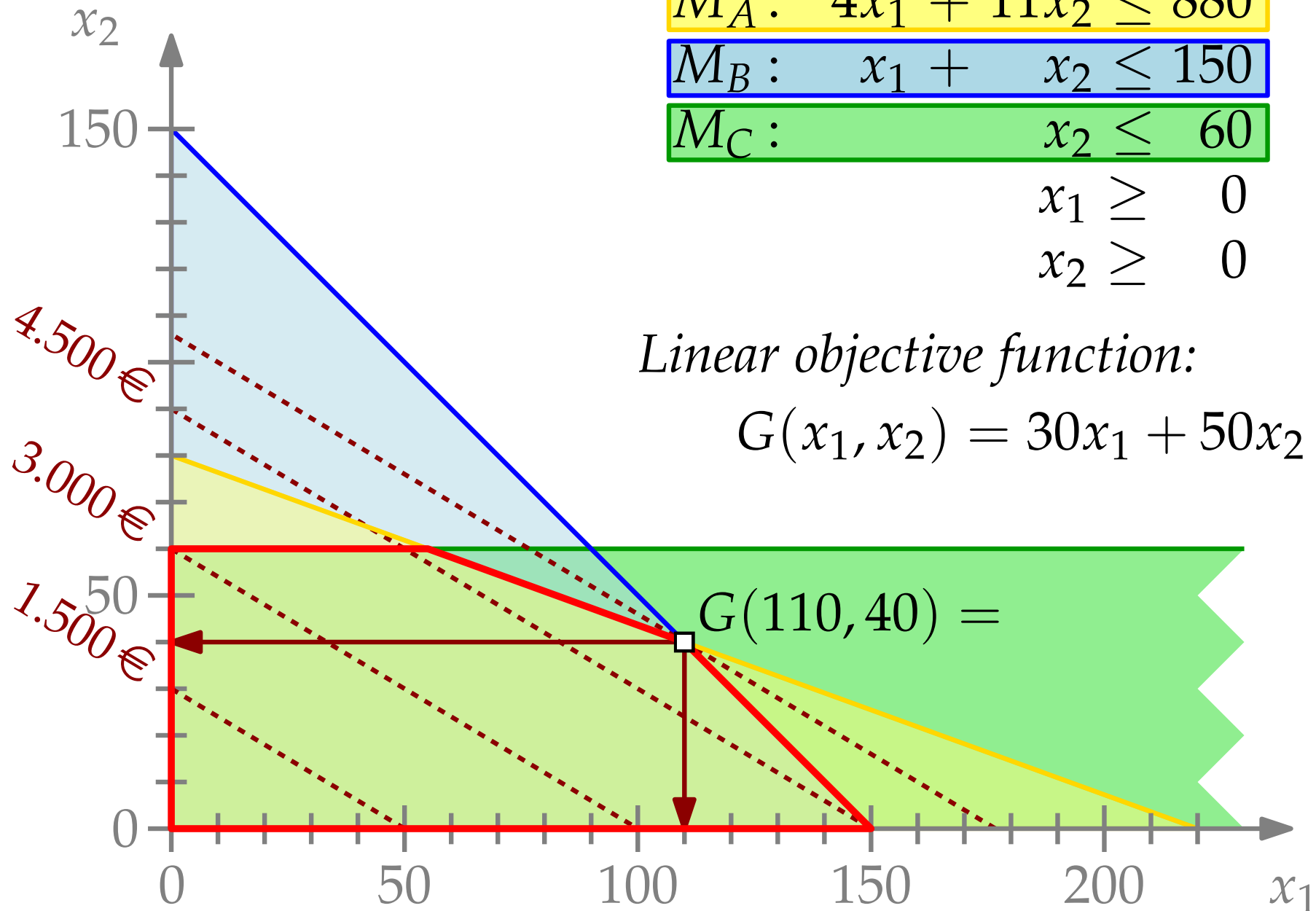
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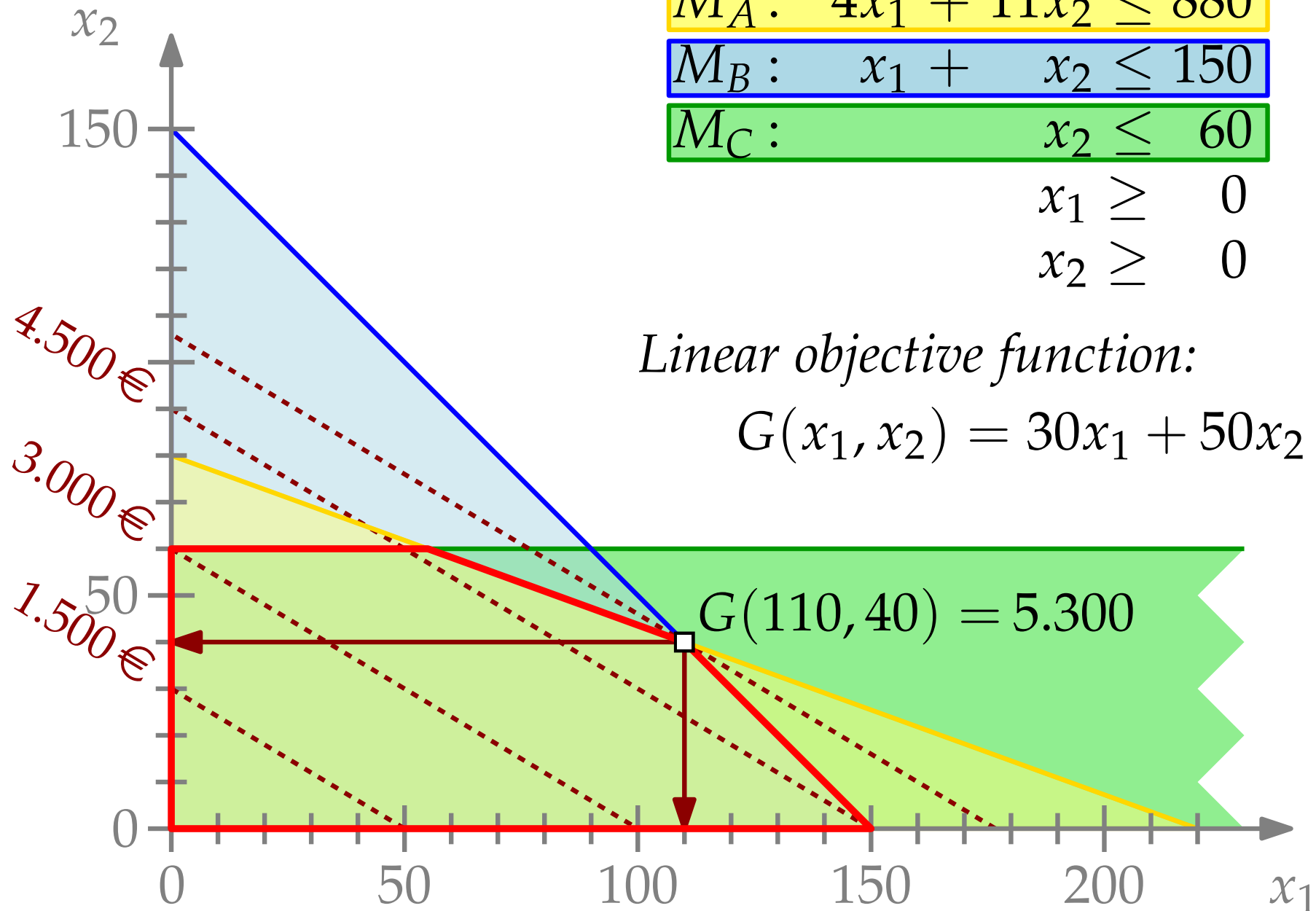
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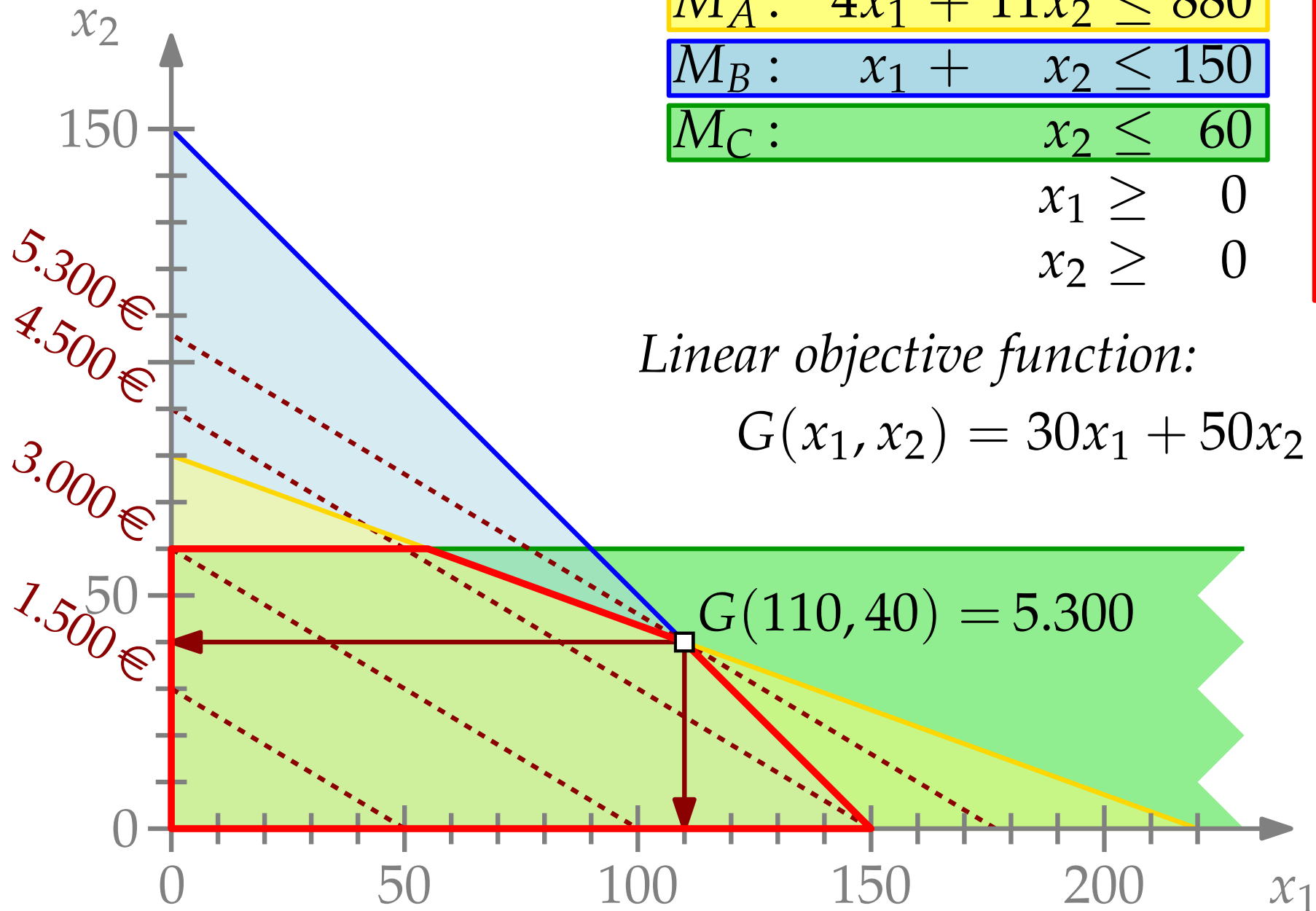
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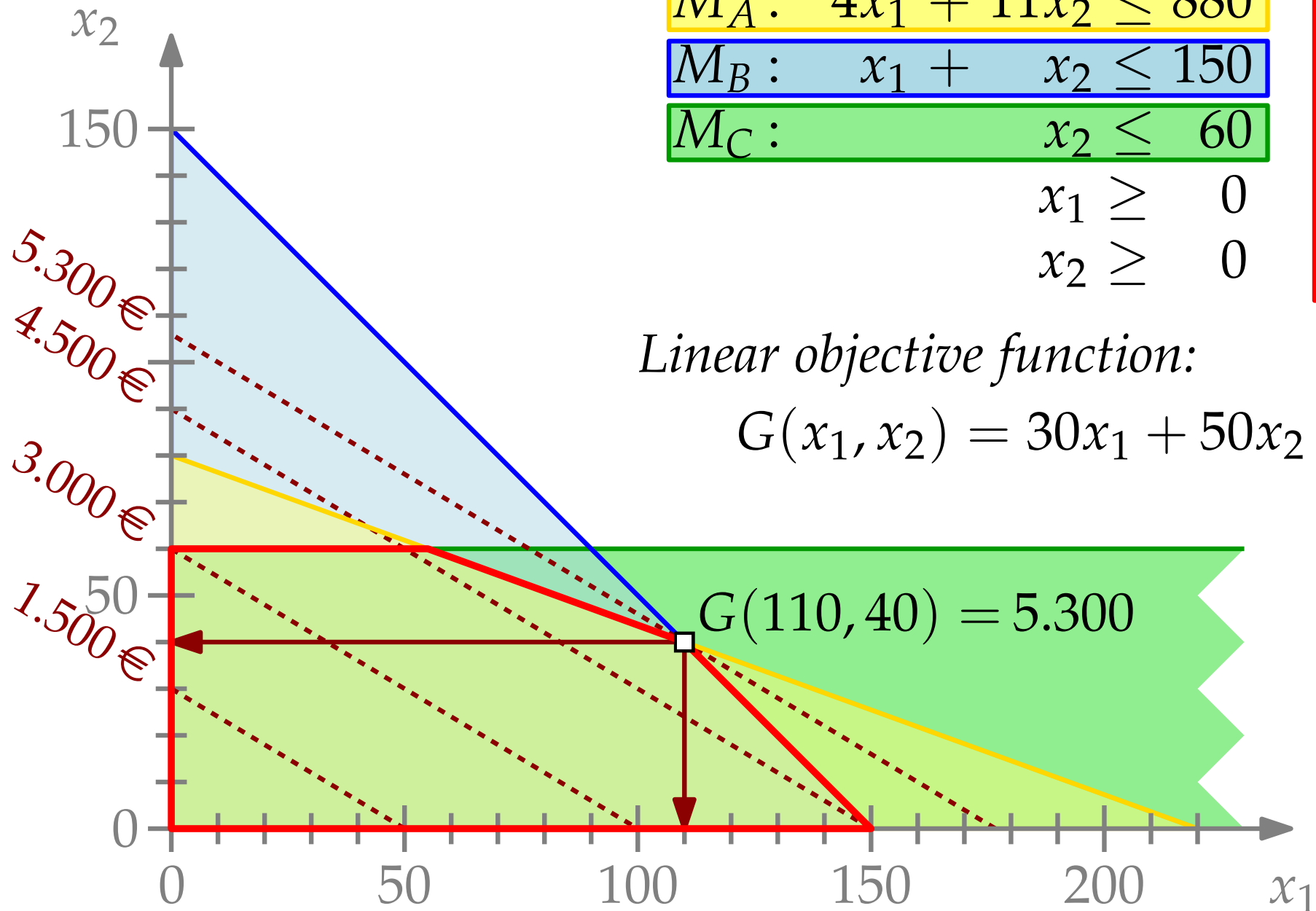
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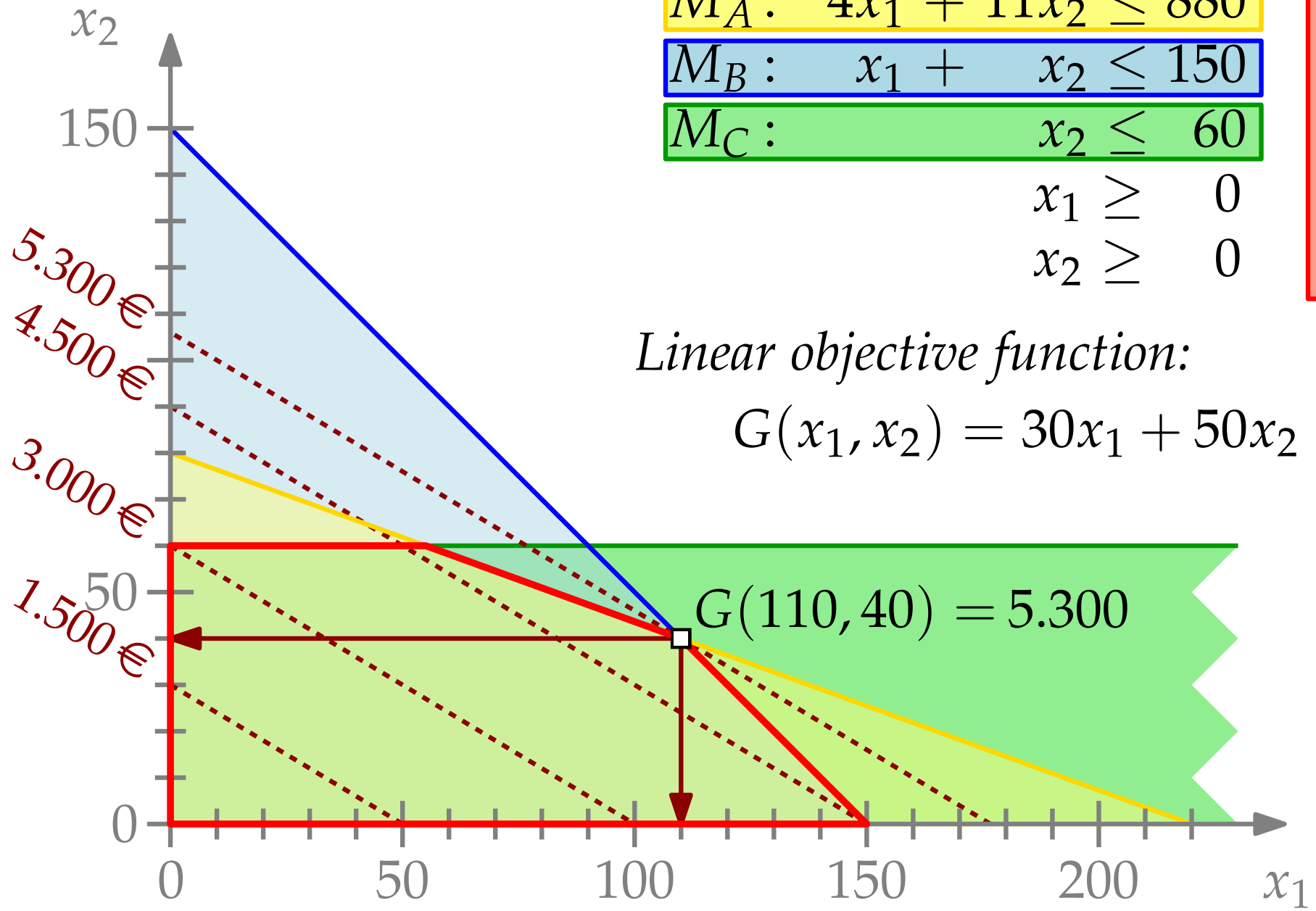
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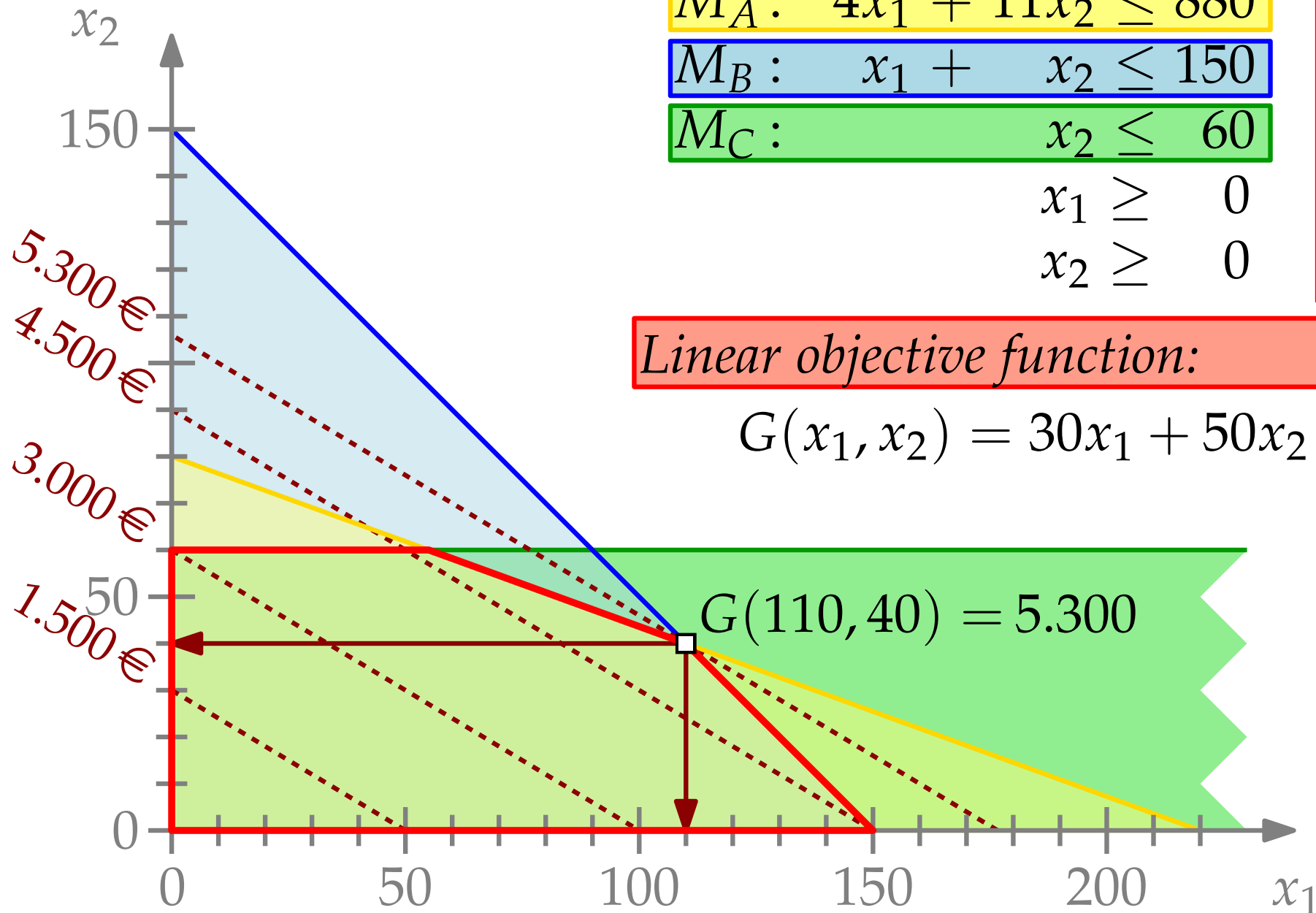
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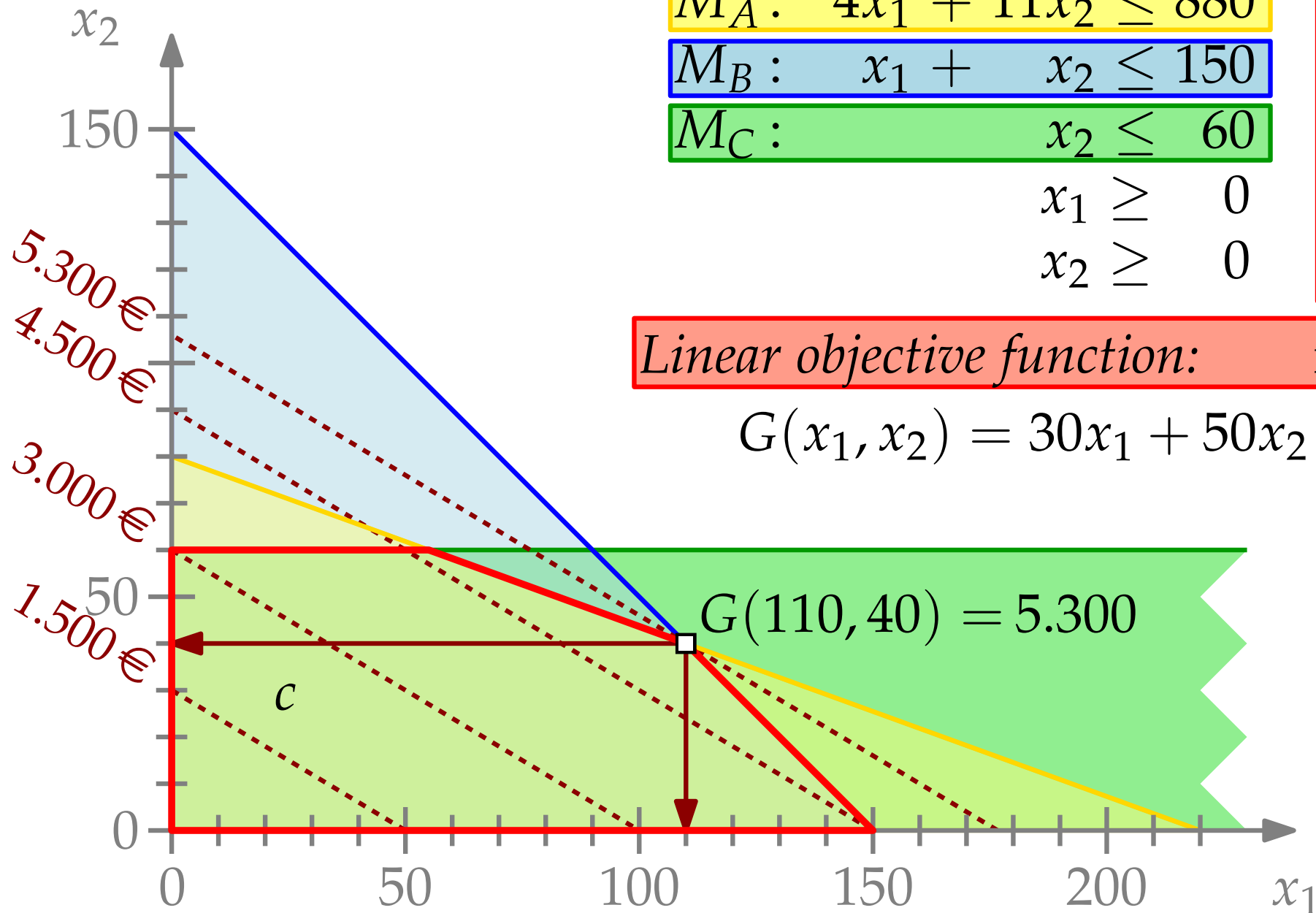
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Linear objective function: $\max c^T x$

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Linear Programming?

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Linear:

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- $x_1 - 4.33x_2 + 0x_4$

Non-Linear:

- $3x_1^2 + 5x_2$
- $10x_1x_2 - 5x_3 + 23 \sin(x_3)$

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- Not programming as in writing code in (say) Java or C++. But it is programming in the sense that we write linear programs (=objective function & constraints) to model and solve real-world problems.

Linear Program (LP)

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$$\begin{aligned} G(x_1, x_2) &= 30x_1 + 50x_2 \\ &= (30, 50) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \end{aligned}$$

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- Sample of hits: “A linear programming approach” ...

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A linear programming approach to the cutting-stock problem

PC Gilmore, RE Gomory - Operations research, 1961 - pubsonline.informs.org

... an approximate solution is being sought by **linear programming**. In this paper, a technique is described for overcoming the difficulty in the **linear programming** formulation of the problem. ...

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A linear programming approach to production and employment scheduling

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[LIBRO] Control of uncertain systems: a linear programming approach

MA Dahleh, IJ Diaz-Bobillo - 1994 - dl.acm.org

We propose a sampling-based **approach** to learn Lyapunov functions for a class of discrete-time autonomous hybrid systems that admit a mixed-integer representation. Such systems ...

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A linear programming approach to max-sum problem: A review

T Werner - IEEE transactions on pattern analysis and machine ..., 2007 - ieeeexplore.ieee.org

... of trees is closest to the **approach** by Schlesinger et al. The **linear programming** relaxation considered by ... The **approach** based on convex combination of trees is more general than the ...

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A linear programming approach to novelty detection

C Campbell, K Bennett - Advances in neural information ..., 2000 - proceedings.neurips.cc

... involves solution of a quadratic **programming** problem. In this paper we propose a simpler kernel method for estimating the support based on **linear programming**. The method is easy to ...

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The linear programming approach to approximate dynamic programming

DP De Farias, B Van Roy - Operations research, 2003 - pubsonline.informs.org

... **linear programming** for approximating solutions to such problems. The **approach** "fits" a **linear** combination of pre-selected basis functions to the dynamic **programming** cost-to-go ...

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An approach to linear programming with 0-1 variables

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... The approximate **approach** using effective gradients for solving problems of **linear programming** with 0-1 variables discussed here will give satisfactory solutions in most cases in a short ...

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Linear programming fundamentals

N Munier - Strategic **Approach** in Multi-Criteria Decision Making: A ..., 2024 - Springer

SIMUS is explained in Chap. 7, and it is the method used in this book to illustrate how most of the aspects related to real-world scenarios can be incorporated in the modelling and ...

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Optimization Study of Cropping Pattern in the Klakah Irrigation Area, Lumajang Regency, Using Linear Programming

W Sejati, TT Akbar - ADI Journal on Recent Innovation, 2024 - adi-journal.org

... The optimization technique used in this crop pattern optimization study employs **linear programming** through the use of the POM QM application. This study plans for 3 alternatives ...

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[PDF] Neutrosophic Fuzzy Regression: A Linear Programming Approach

M Darehmiraki - Iranian Journal of Operations Research, 2024 - iors.ir

... This work introduces a **linear programming approach** for constructing regression models for ... deviation **approach** to transform the regression issue into a **linear programming** problem. In ...

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Linear program for testing nonclassicality and an open-source implementation

JH Selby, E Wolfe, D Schmid, AB Sainz, VP Rossi - Physical Review Letters, 2024 - APS

... Indeed, the **linear program** we derive is simply a test of whether ... -source code implements the **linear program** we develop herein. ... For now, we simply note that the **linear program** (and ...

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[HTML] Wave energy site location optimizing in Chile: A fuzzy serial linear programming decision-making approach

CN Wang, NL Nhieu, HV Tran - Environment, Development and ..., 2024 - Springer

... To narrow this practice research gap, this study proposes hybrid **approaches** between DEA, FBWM and Bonferroni FCoCoSo to determine suitable locations for WESS. ...

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Improving estimates accuracy of voter transitions. Two new algorithms for ecological inference based on linear programming

JM Pavia, R Romero - Sociological Methods & Research, 2024 - journals.sagepub.com

... First, we suggest a novel procedure, based on **linear programming** and grounded in the homogeneity hypothesis, to estimate the inner-cells values at the local level. We call this ...

☆ Salva 99 Cita Citato da 16 Articoli correlati Tutte e 8 le versioni Web of Science: 4

Hybrid quantum-classical computing via Dantzig-Wolfe decomposition for integer linear programming

X Wei, J Liu, L Fan, Y Guo, Z Han - ..., 2024 - ieeeexplore.ieee.org

... 1, in HyDWD, the master problem constitutes a **linear programming** problem solvable on a ... In Section III, we briefly review the classical integer **linear programming** problem and its ...

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Linear Program (LP)

Given: matrix $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$

Find: $x^* \in \mathbb{R}^n$ where $x^* = \arg \max \{c^T x \mid Ax \leq b\}$

- Software that solves linear programs – *linear programming solvers* – also generate lots of important auxiliary information (as well as the optimum).
- Sensitivity analysis, shadow prices, alternative optima (all very useful for understanding the meaning, stability and significance of the optimum solutions generated by linear programming solvers; important in industry.)

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- ok, ok, but how do we solve LPs ?

Why study the Math of LPs?

- Many people can do basic modelling using linear programming. In many cases this is good enough.
- But the separation of what and how is, of course, still not complete.
- ...so your job is to also understand what is going on “in the engine room”
- This will help you get maximum performance from your LP solver and deal with phenomena such as degeneracy.

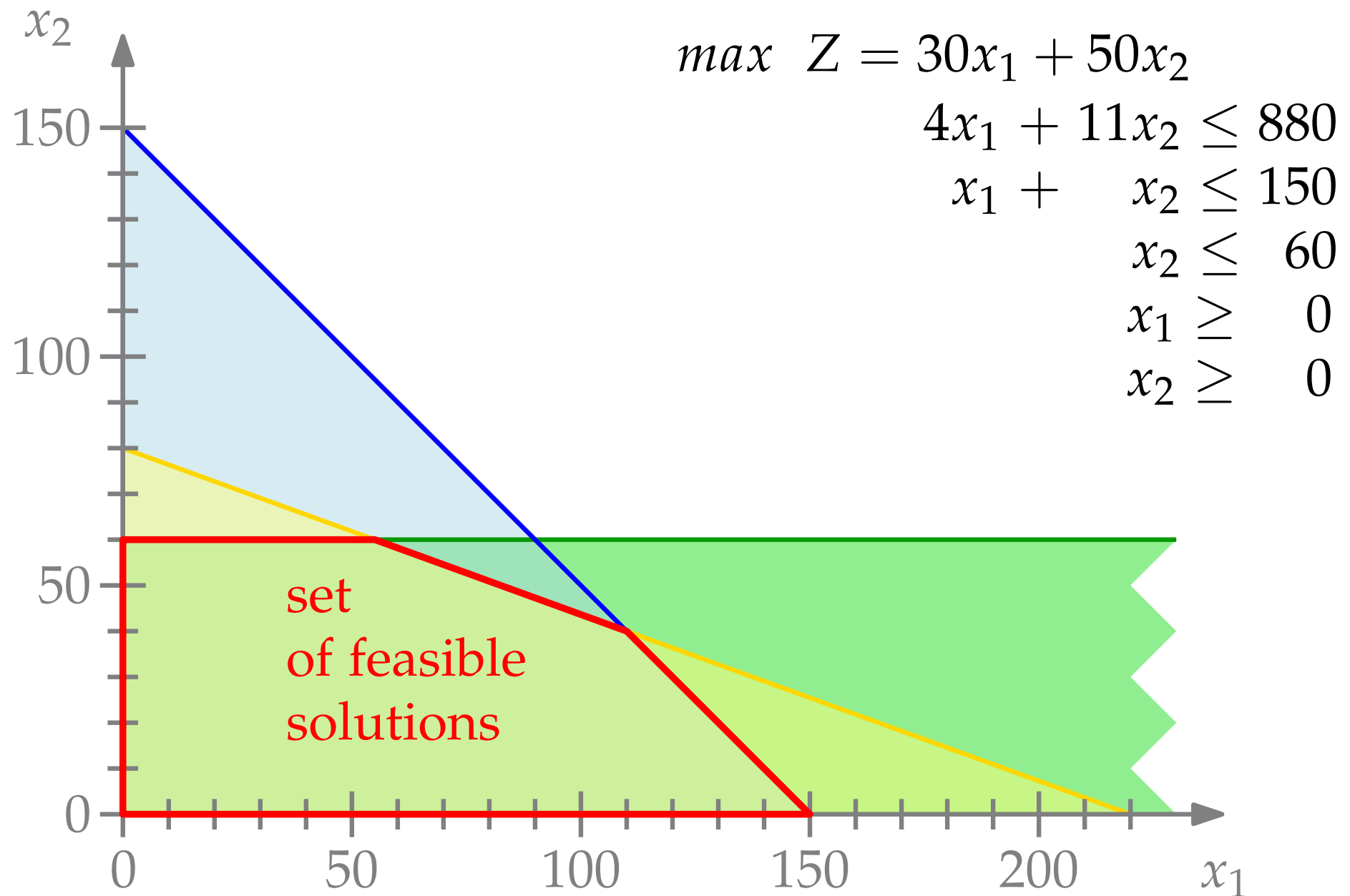
Why study the Math of LPs?

- LP can be viewed as the first step in a hierarchy of increasingly complex forms of mathematical programming (non-linear programming, especially convex programming; integer programming)
- Understanding the mathematics of LP is an important step towards understanding these mathematically more complex paradigms
- The abstract mathematical theory behind linear programming turns out to be extremely important for certain theoretical mathematical fields, e.g., approximation algorithms built around the scaffolding of LP- duality.
- Many of these scientists will never write or solve linear programs, but the theory of linear programming will be ever-present in their work.

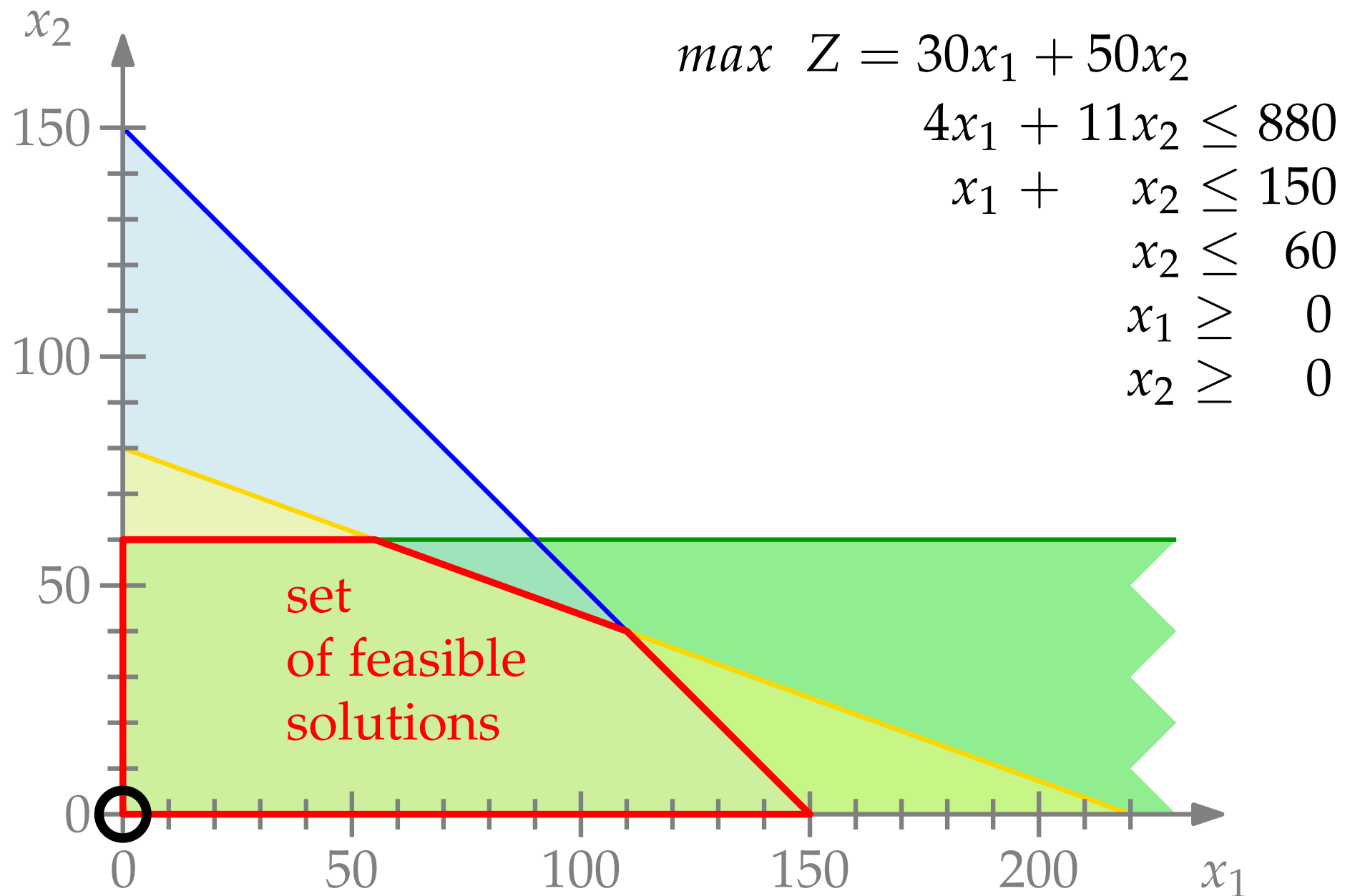
Solving LPs

- Most linear programming solvers are based on either or both of the following two techniques:
 - The Simplex method
 - Interior-point methods
- It is not trivial to solve really big linear programs. All kinds of extra mathematical tricks are needed as the size of the program grows.

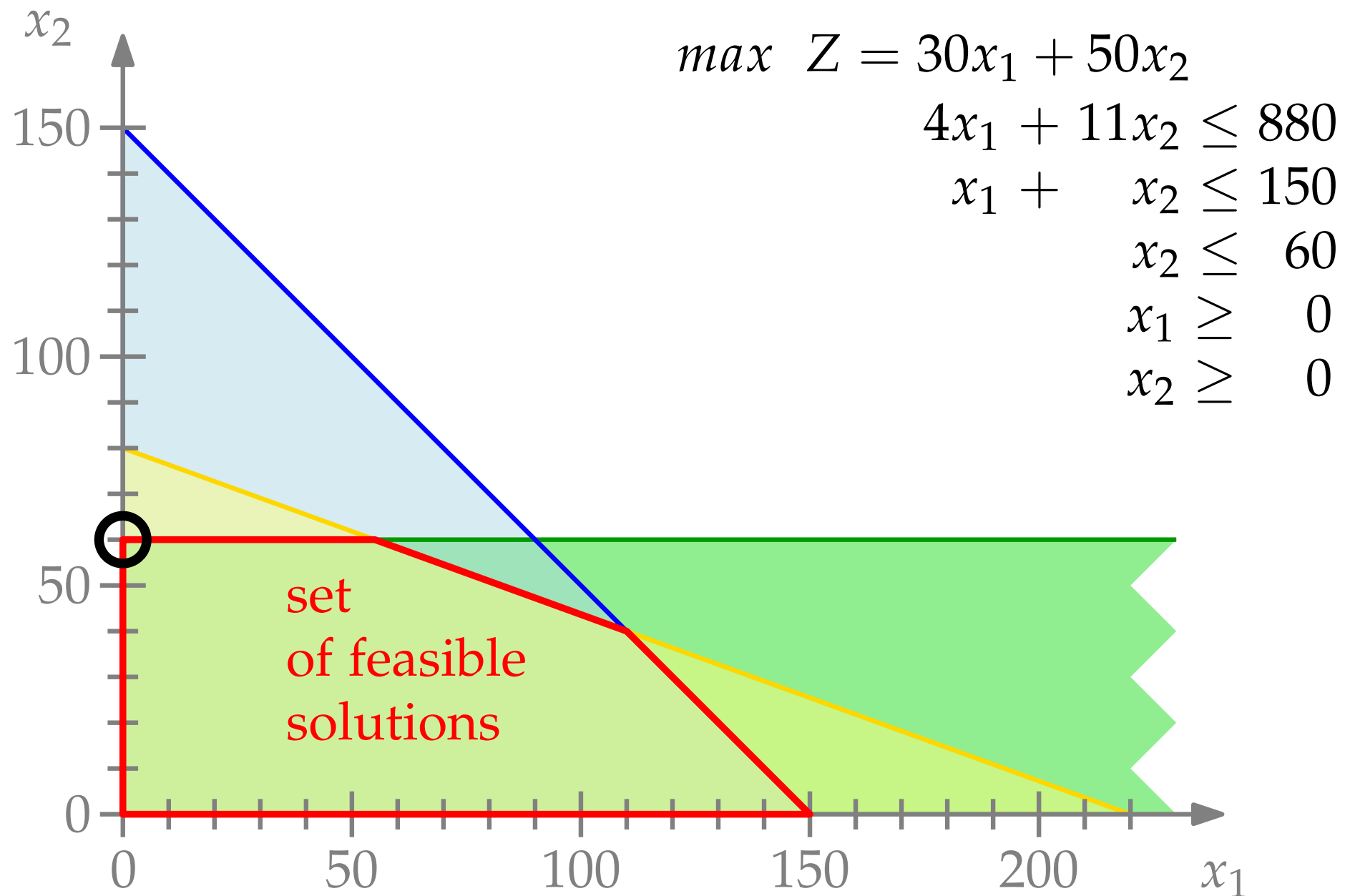
Exterior Point Methods



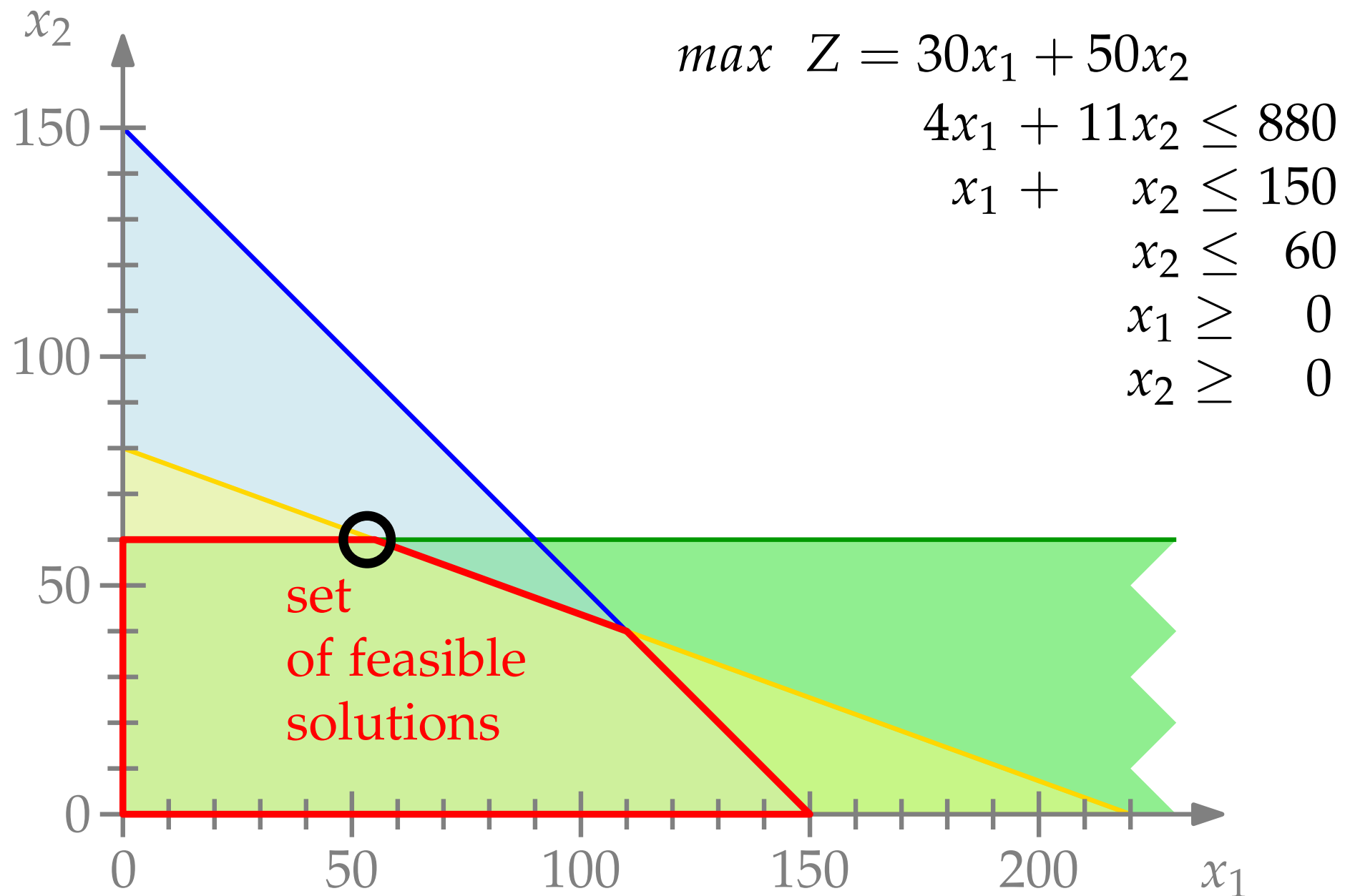
Exterior Point Methods



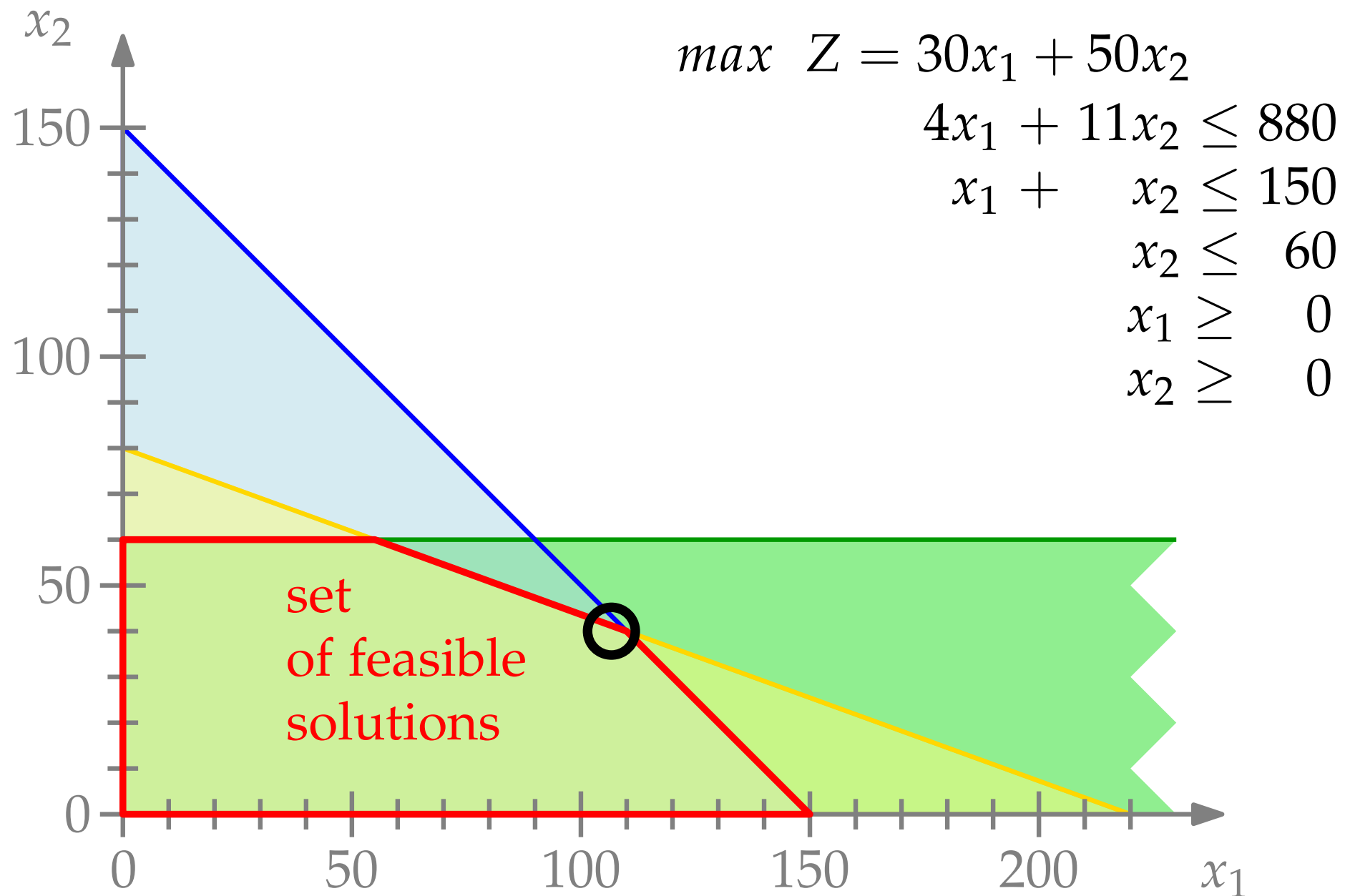
Exterior Point Methods



Exterior Point Methods



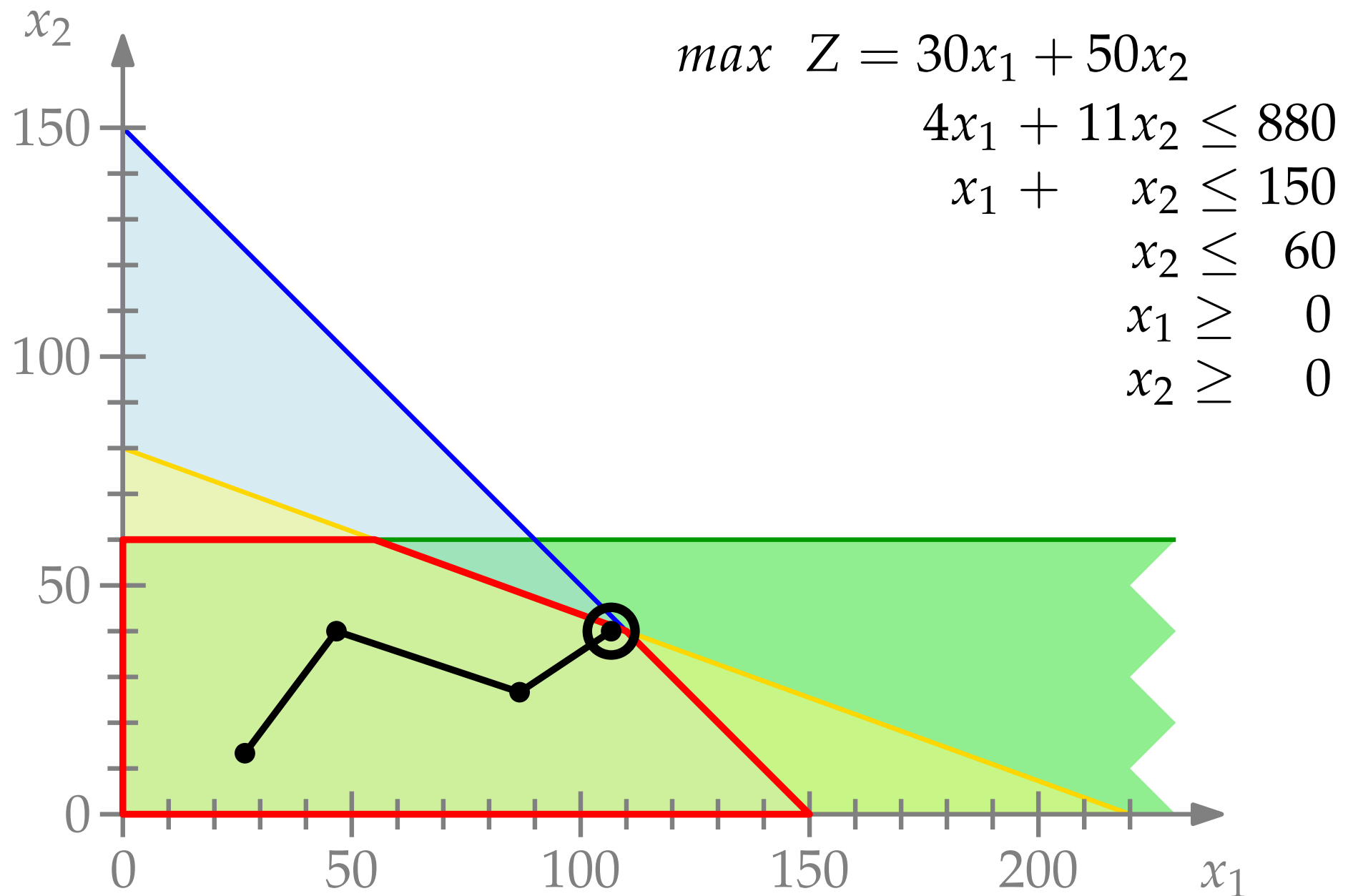
Exterior Point Methods



The Simplex method

- The most frequently used algorithm to solve linear programs
- The core algorithm is quite simple (...)
- Exterior-point method: it goes around the perimeter of the feasible region, one cornerpoint at a time
- Very useful for extended analysis: re- optimisation, shadow prices, sensitivity analysis etc.
- “Bad cases” for the algorithm hardly ever occur in practice

Interior Point Methods



Interior Points Method

- More recent than the Simplex method
- These methods go through the middle of the feasible region, not around the outside
- Per iteration, much slower than the Simplex method
But the number of iterations required increases much more slowly than with the Simplex method
- Thus: superior/competitive for extremely large instances

Terminology

Barbara Franci

The Wyndor Glass Example

■ **TABLE 3.1** Data for the Wyndor Glass Co. problem

Plant	Production Time per Batch, Hours		Production Time Available per Week, Hours
	Product		
	1	2	
1	1	0	4
2	0	2	12
3	3	2	18
Profit per batch	\$3,000	\$5,000	

The Wyndor Glass Example

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$$\begin{aligned}
 &\text{Maximize} && Z = && 3x_1 &+& 5x_2 \\
 &\text{subject to} && && x_1 && \leq 4 \\
 &&& && && x_2 \leq 12 \\
 &&& && 3x_1 &+& 2x_2 \leq 18 \\
 &\text{and} && x_1 \geq 0, && x_2 \geq 0
 \end{aligned}$$

The Wyndor Glass Example

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*Objective
Function*

Maximize $Z = 3x_1 + 5x_2$

subject to

$$\begin{aligned} x_1 &\leq 4 \\ x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \end{aligned}$$

and $x_1 \geq 0, x_2 \geq 0$

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x_1 and x_2 :
Decision Variables

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 x_2 &\leq 12 \\
 3x_1 + 2x_2 &\leq 18
 \end{aligned}$$
Functional Constraints

Decision Variables x_1 and x_2 : and $x_1 \geq 0, x_2 \geq 0$

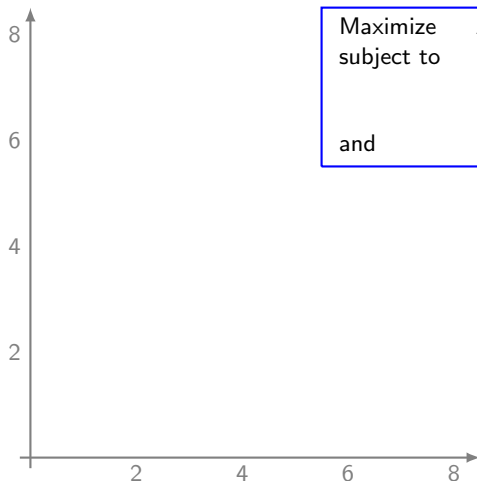
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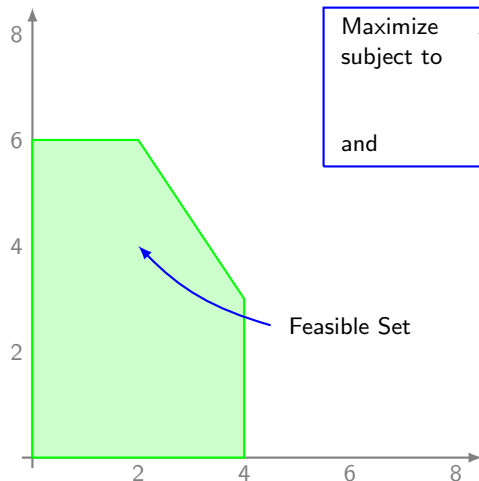
Objective Function *Functional Constraints*
 Maximize $Z = 3x_1 + 5x_2$
 subject to
 $x_1 \leq 4$
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 $3x_1 + 2x_2 \leq 18$
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Nonnegativity Constraints
 x_1 and x_2 : *Decision Variables*

The Wyndor Glass Example



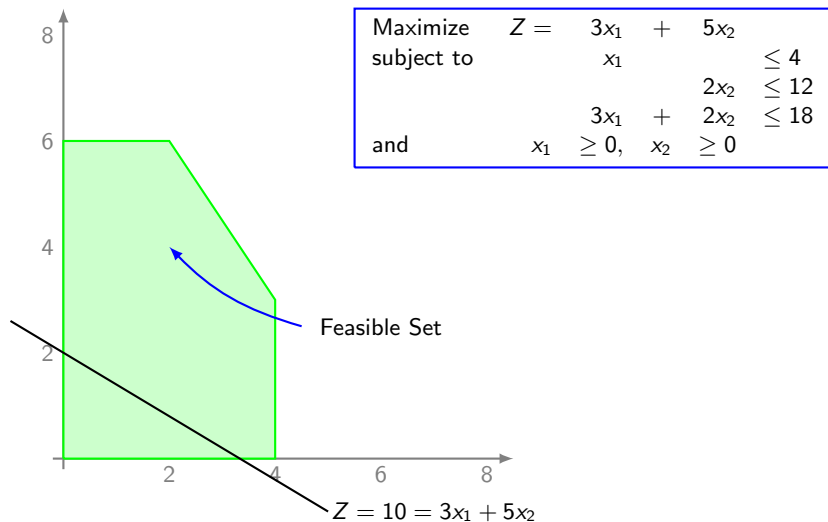
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The Wyndor Glass Example

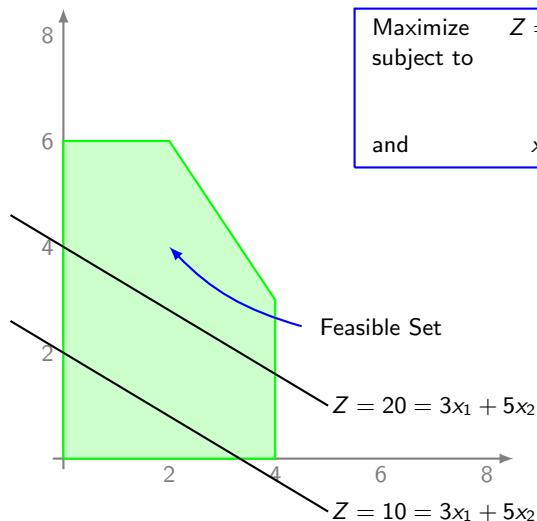


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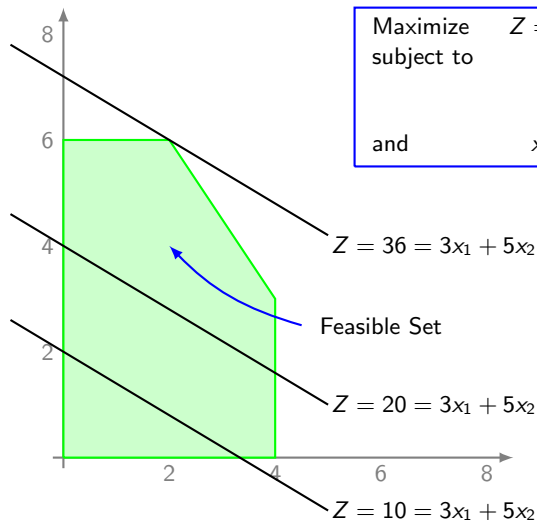


The Wyndor Glass Example



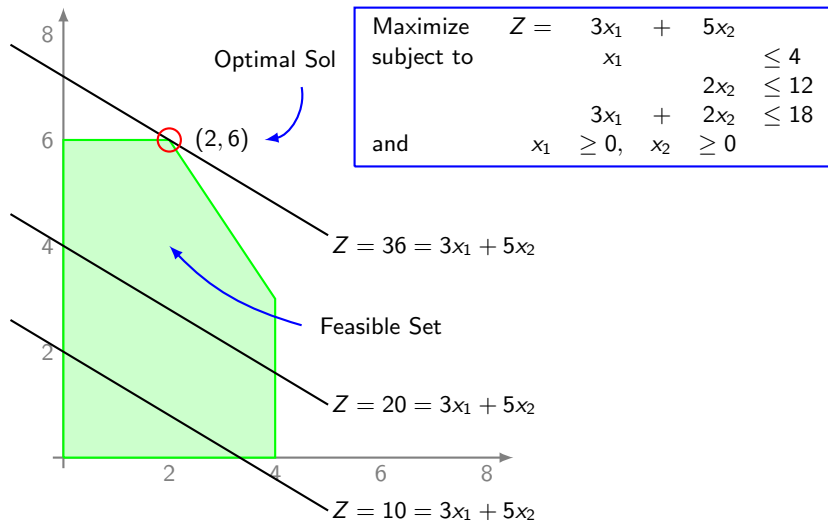
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The Wyndor Glass Example



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The Wyndor Glass Example

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The Wyndor Glass Example

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- ▶ However, it can happen that the feasible region is **empty**, i.e., there are no solutions possible. We then say that the linear program is **infeasible**.

The Wyndor Glass Example

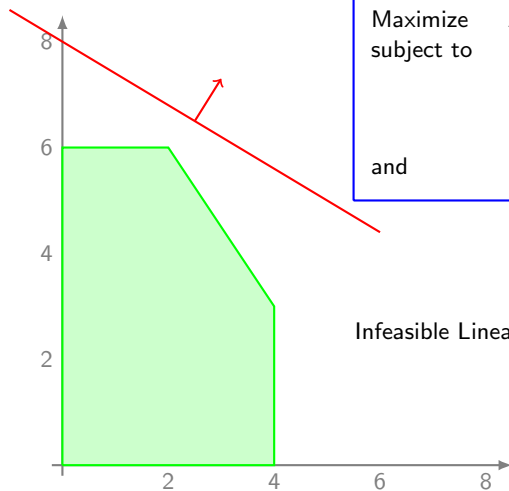
- ▶ In this example, the feasible region is **non-empty**, and **bounded**.
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- ▶ There can also be **multiple optima**.

The Wyndor Glass Example

Infeasible

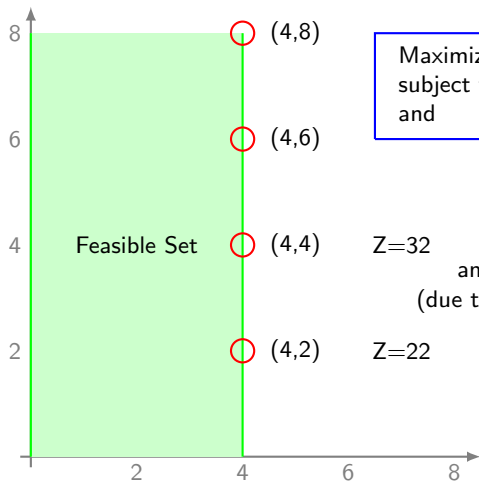


$$\begin{array}{llllll} \text{Maximize} & Z = & 3x_1 & + & 5x_2 & \\ \text{subject to} & & x_1 & & & \leq 4 \\ & & & & x_2 & \leq 12 \\ & & 3x_1 & + & 2x_2 & \leq 18 \\ & & 3x_1 & + & 5x_2 & \geq 40 \\ \text{and} & & x_1 & \geq 0, & x_2 & \geq 0 \end{array}$$

Infeasible Linear Program

The Wyndor Glass Example

Unbounded



$$\begin{array}{ll} \text{Maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \leq 4 \\ \text{and} & x_1 \geq 0, \quad x_2 \geq 0 \end{array}$$

$Z=32$ Unbounded Feasible Set
and Unbounded Objective Function
(due to the angle of the objective function)

$Z=22$

The Wyndor Glass Example

Multiple Optima

